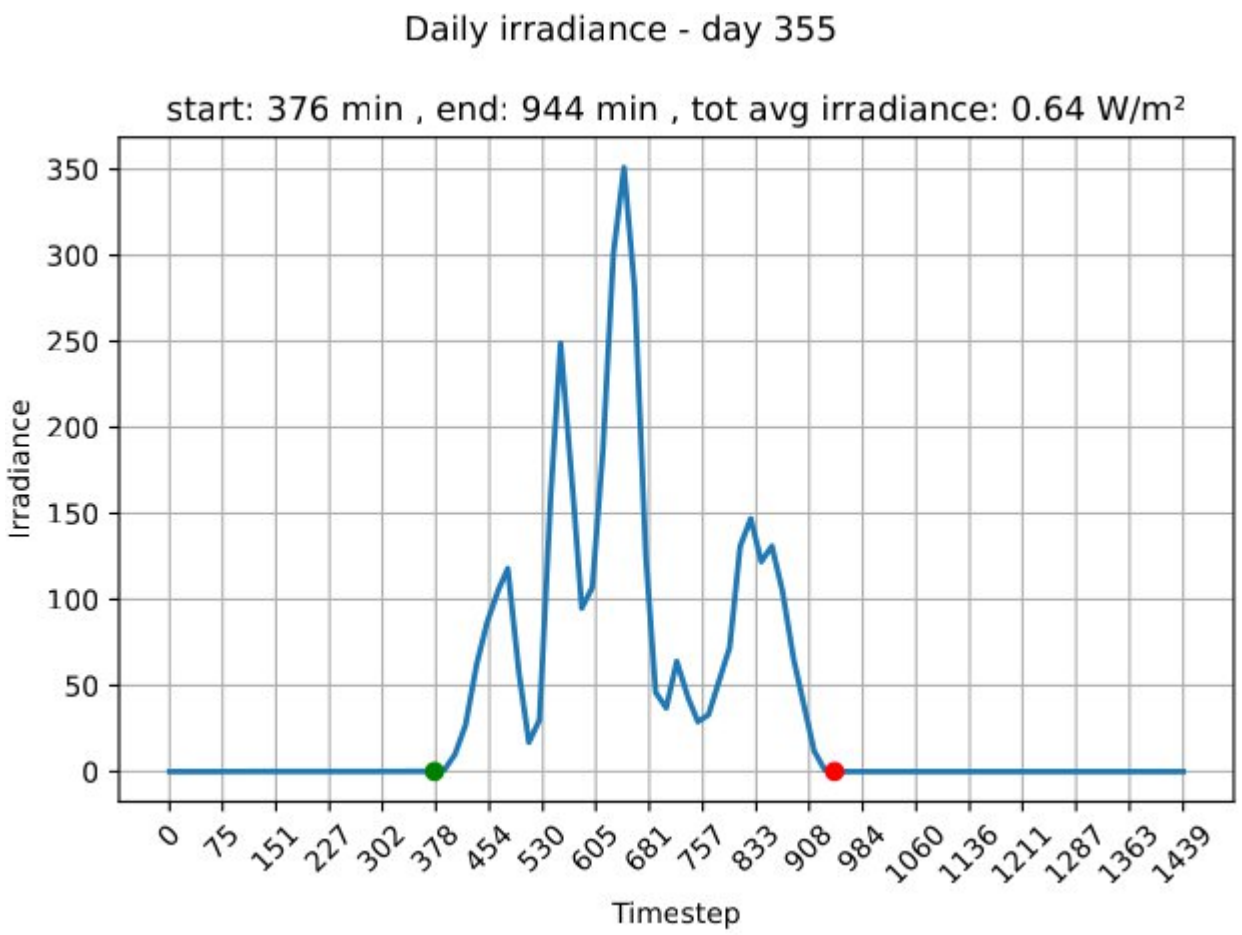
MARL: Tabular vs DQN

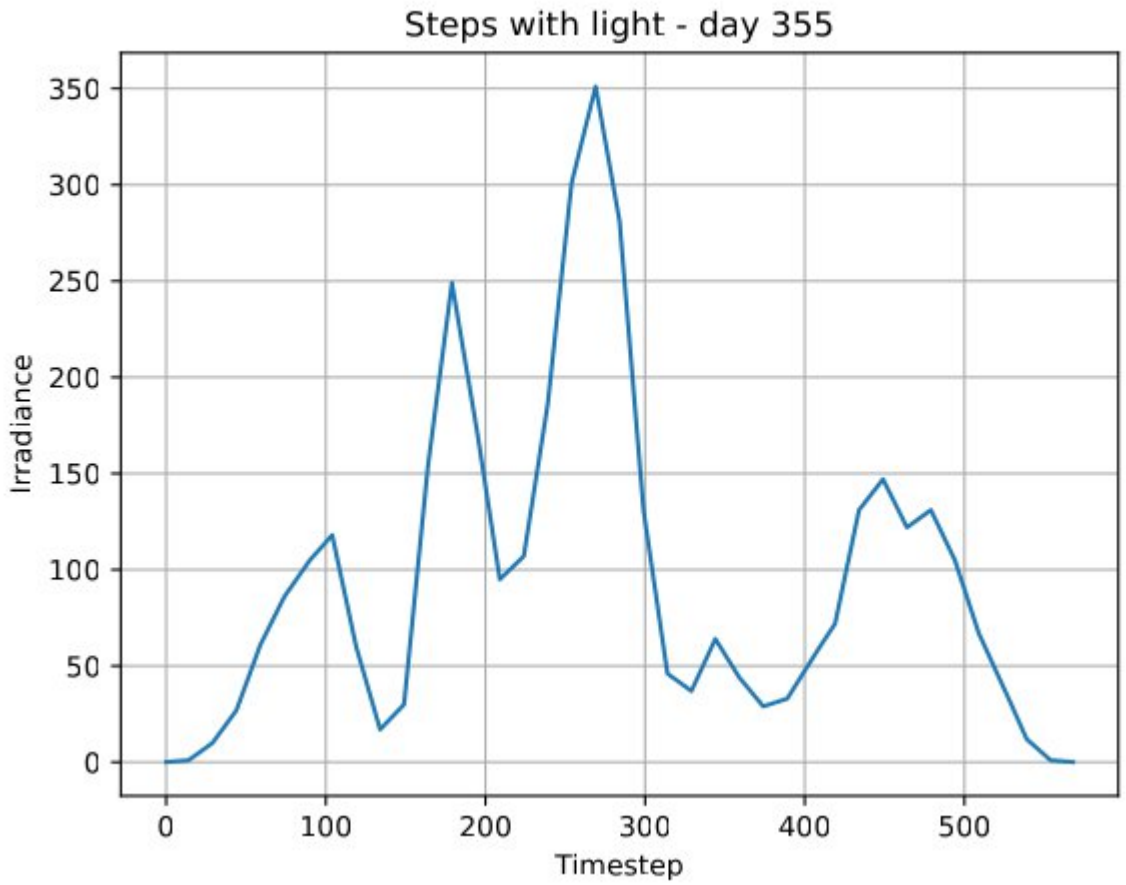
Lorenzo Pecorari - 1885161

Setup

- 2 nodes, both as an agent
- Battery capacities: 25Wh, 100Wh
- PV surfaces: 1.0 m², 0.5 m² (efficiency = 20% for both of them)
- Dataset from Solcast, year 2024
- Observation space: $s_t = [s_t^0, s_t^1]$, where $s_t^i = [b_t^i, \sigma_t^i, \tau_t^i]$
- Action space: $a_t = [a_t^0, a_t^1]$, where $a_t^i = [f_t^i, x_t^i, g_t^i, h_t^i]$
- Each agent can see only the state of the others but not their actions
- Same environment and same reward function for both implementations
- Tabular uses matrix while DQN uses a customization of Stable Baselines 3

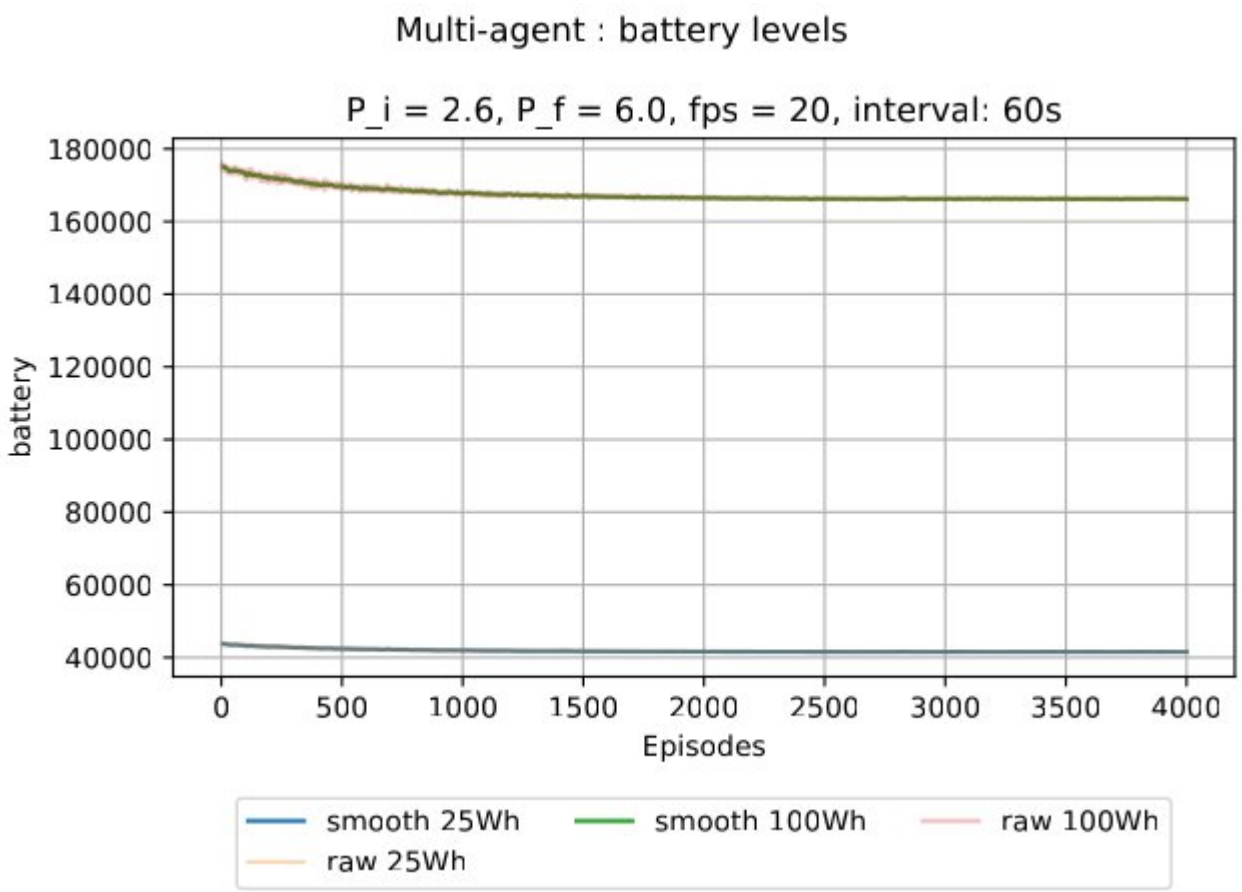
Day 355


Daily irradiance - day 355
start: 376 min , end: 944 min , tot avg irradiance: 0.64 W/m²
Irradiance
Timestep
0
50
100
150
200
250
300
350
0
75
151
227
302
378
454
530
605
681
757
833
908
984
1060
1136
1211
1287
1363
1439

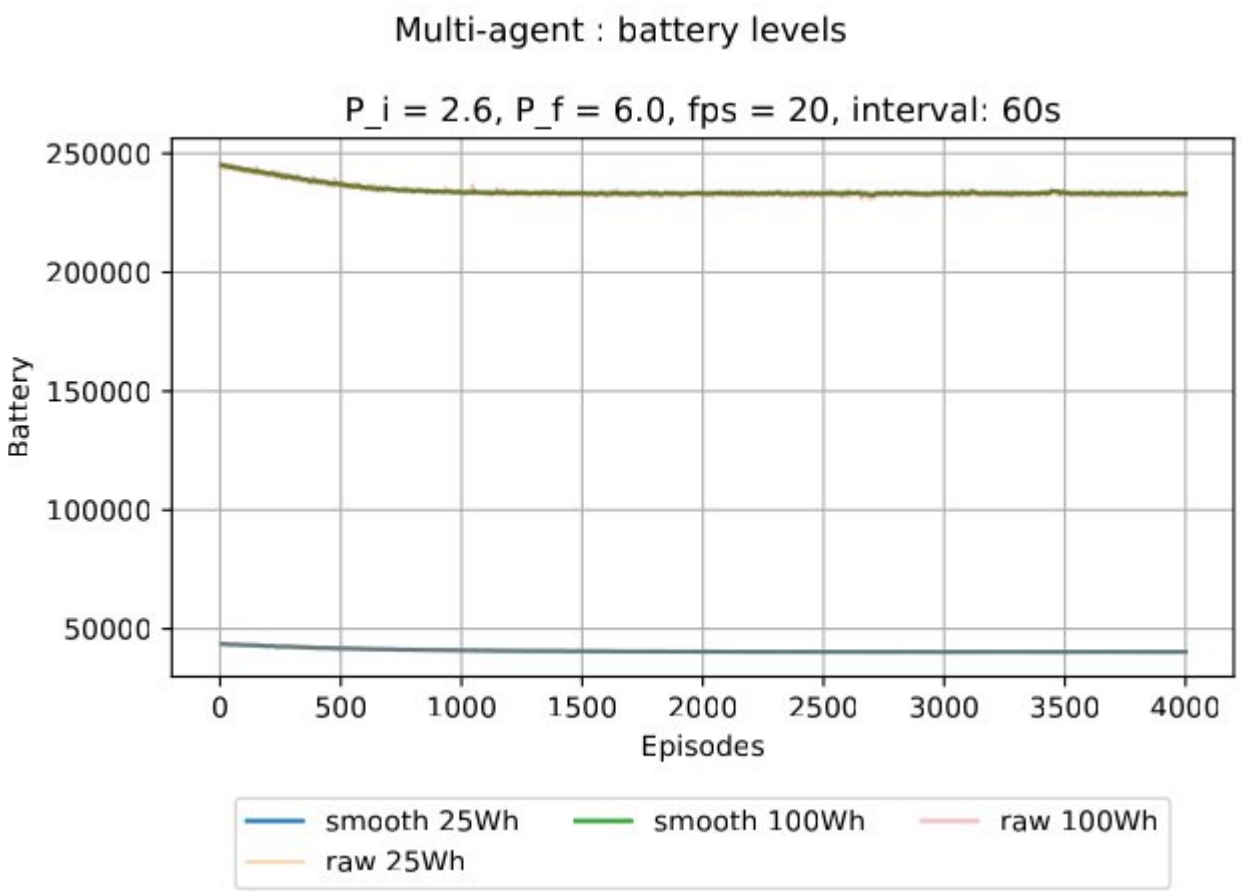

Steps with light - day 355
Irradiance
Timestep
0
50
100
150
200
250
300
350
0
100
200
300
400
500

Day 355: batteries

TABULAR

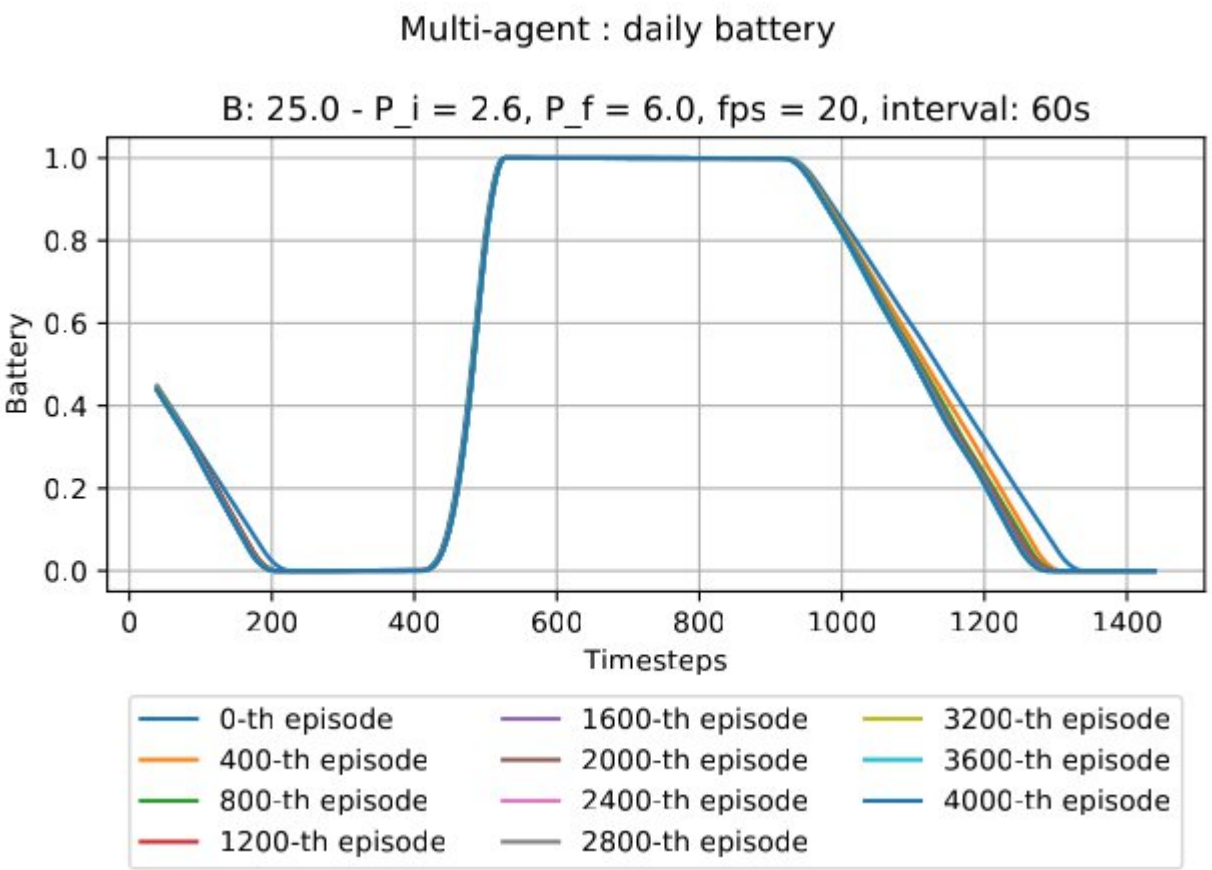


DQN

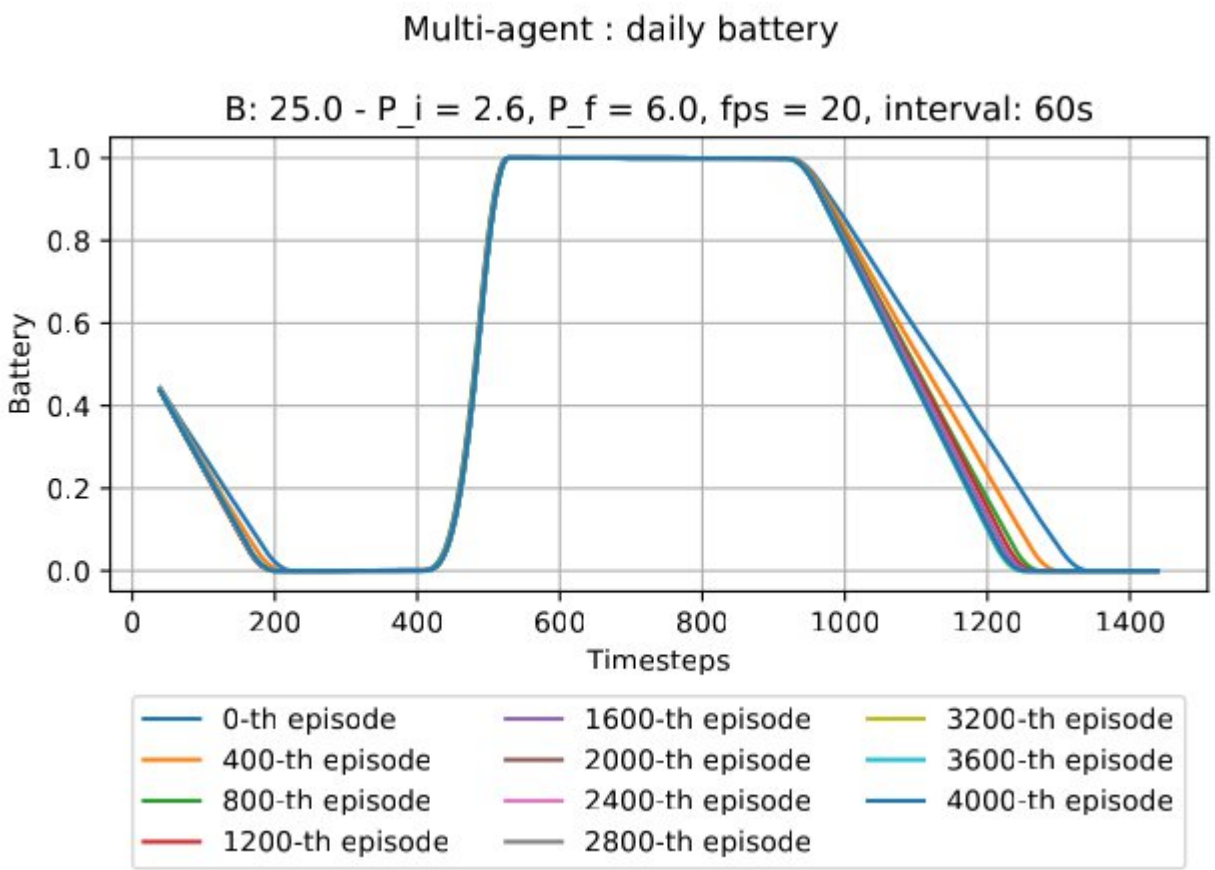


Day 355: battery 25Wh

TABULAR

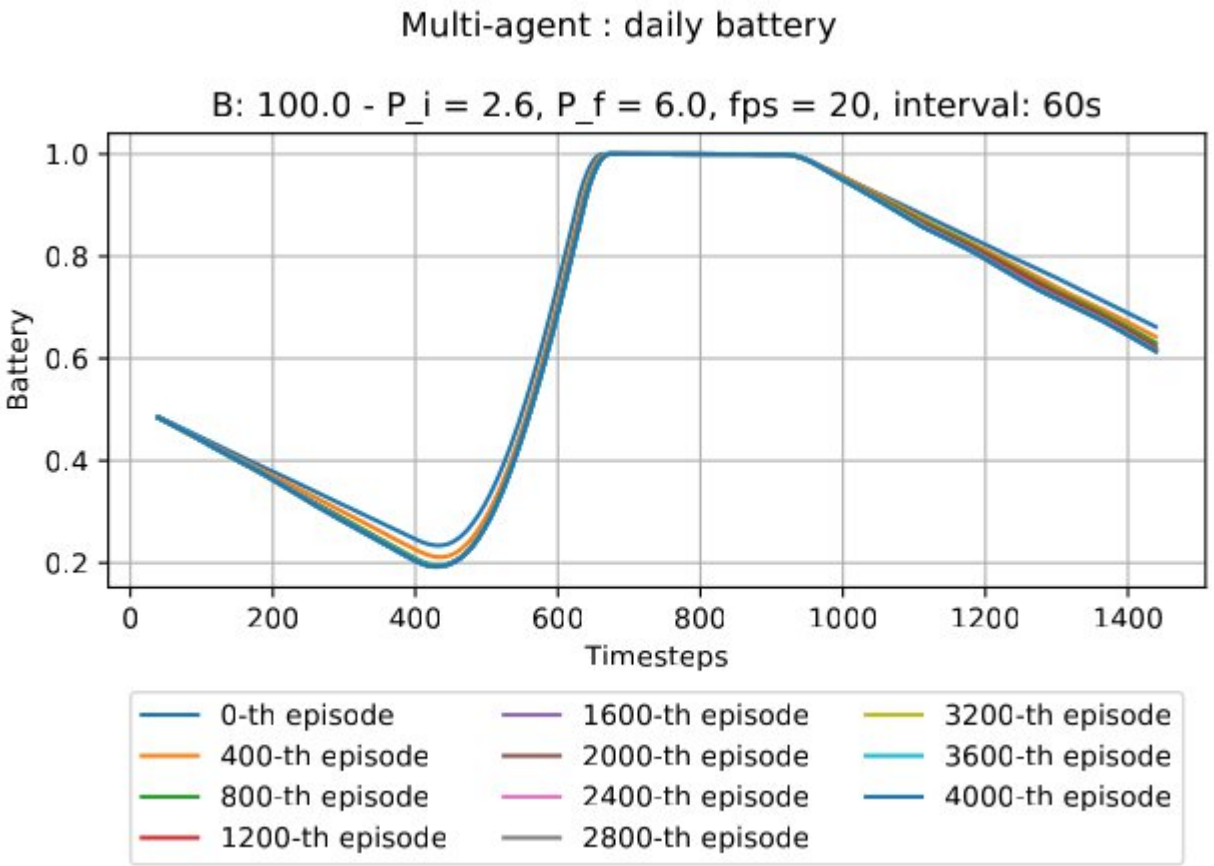


DQN



Day 355: battery 100Wh

TABULAR



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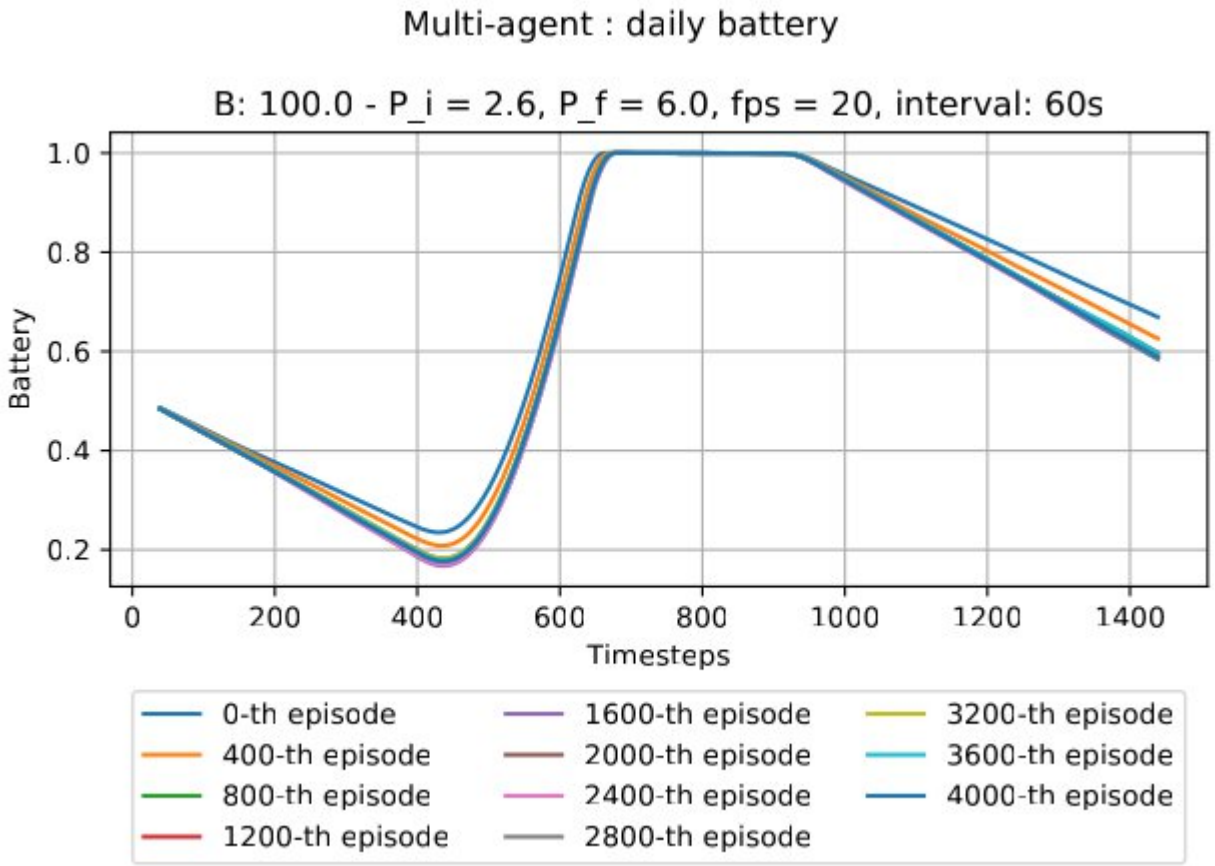
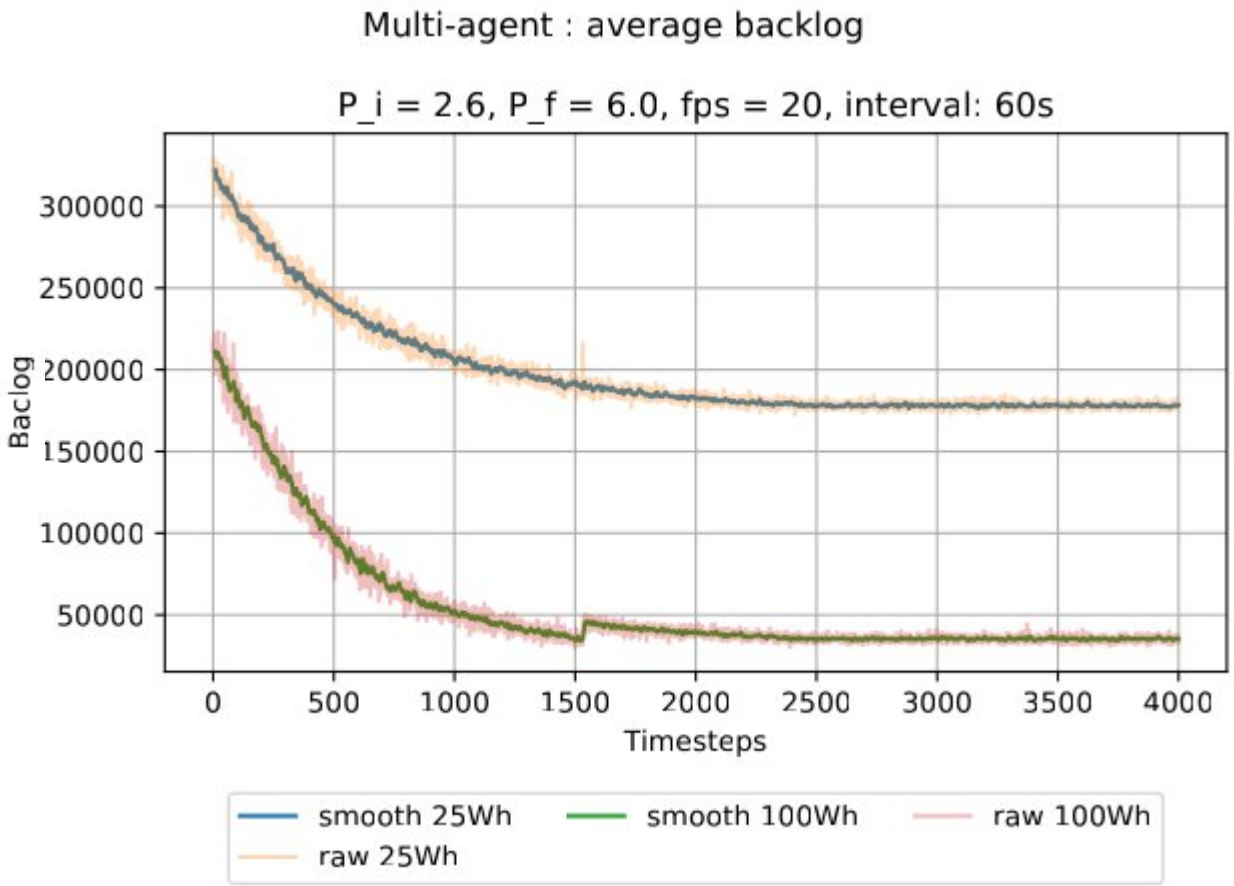
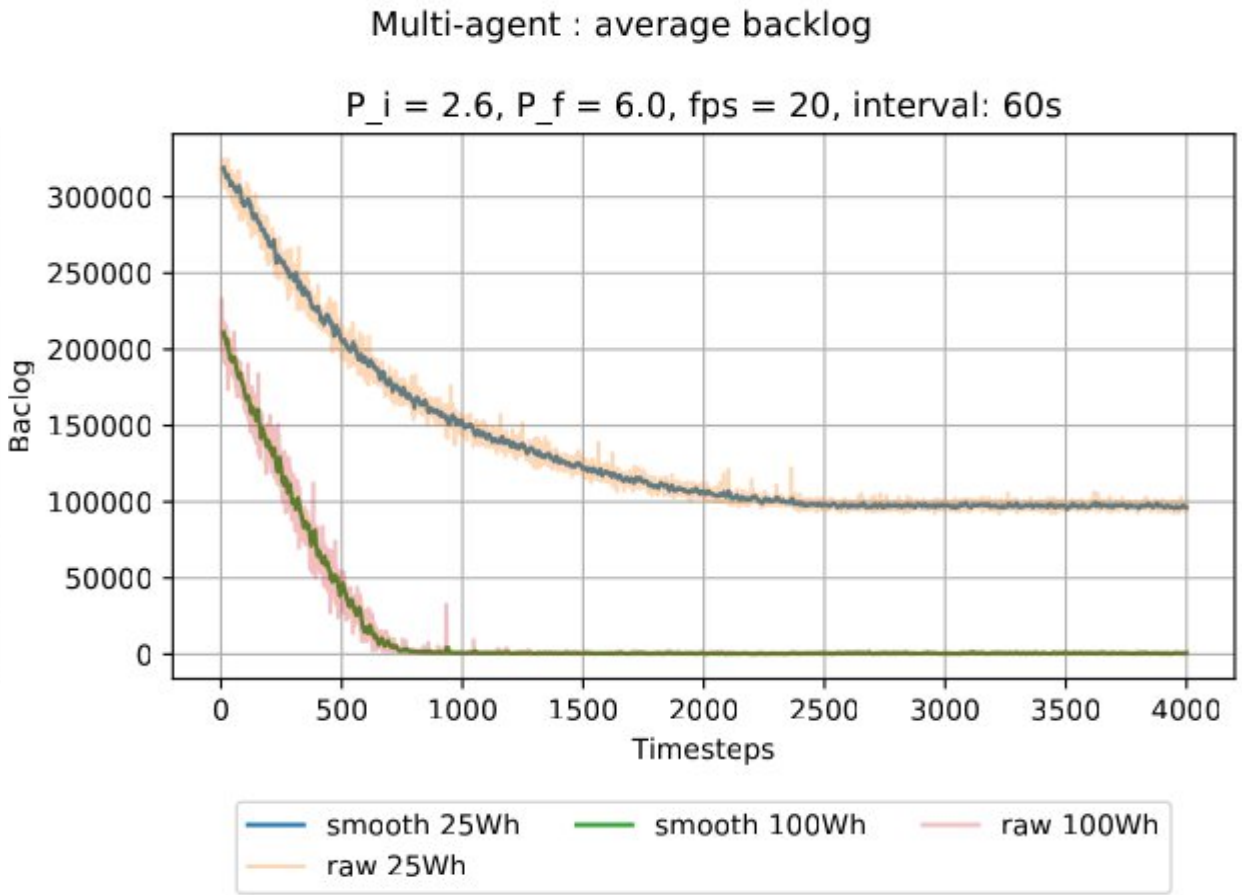


Day 355: backlogs

TABULAR

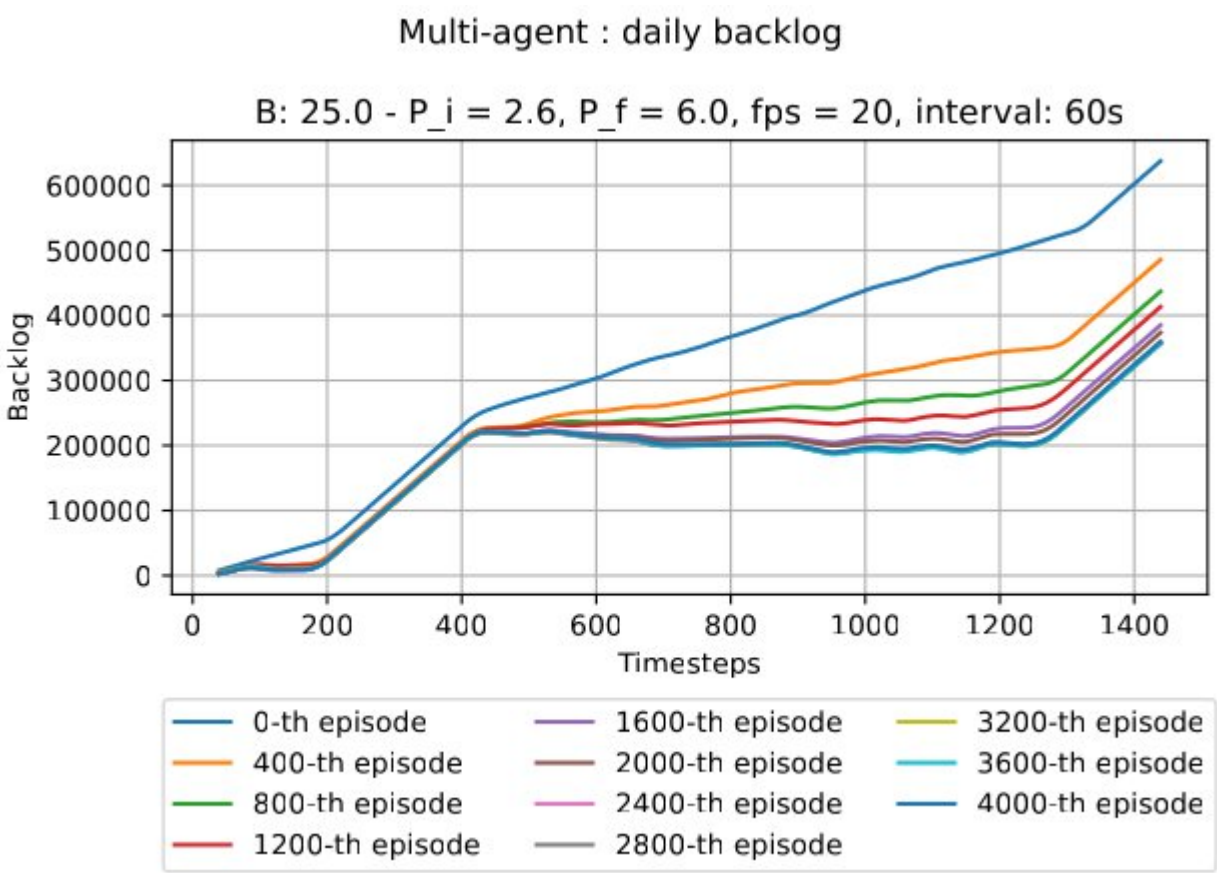


DQN

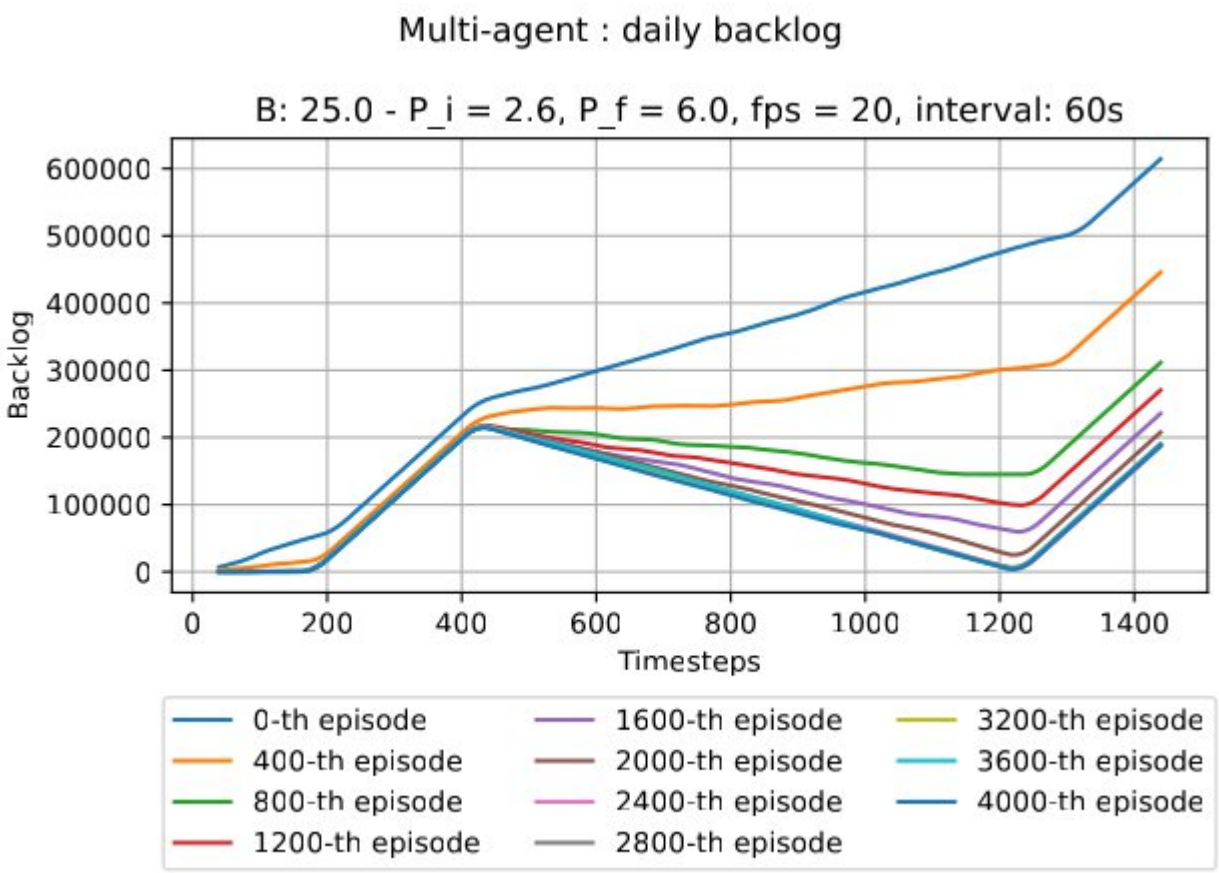


Day 355: backlog 25Wh

TABULAR

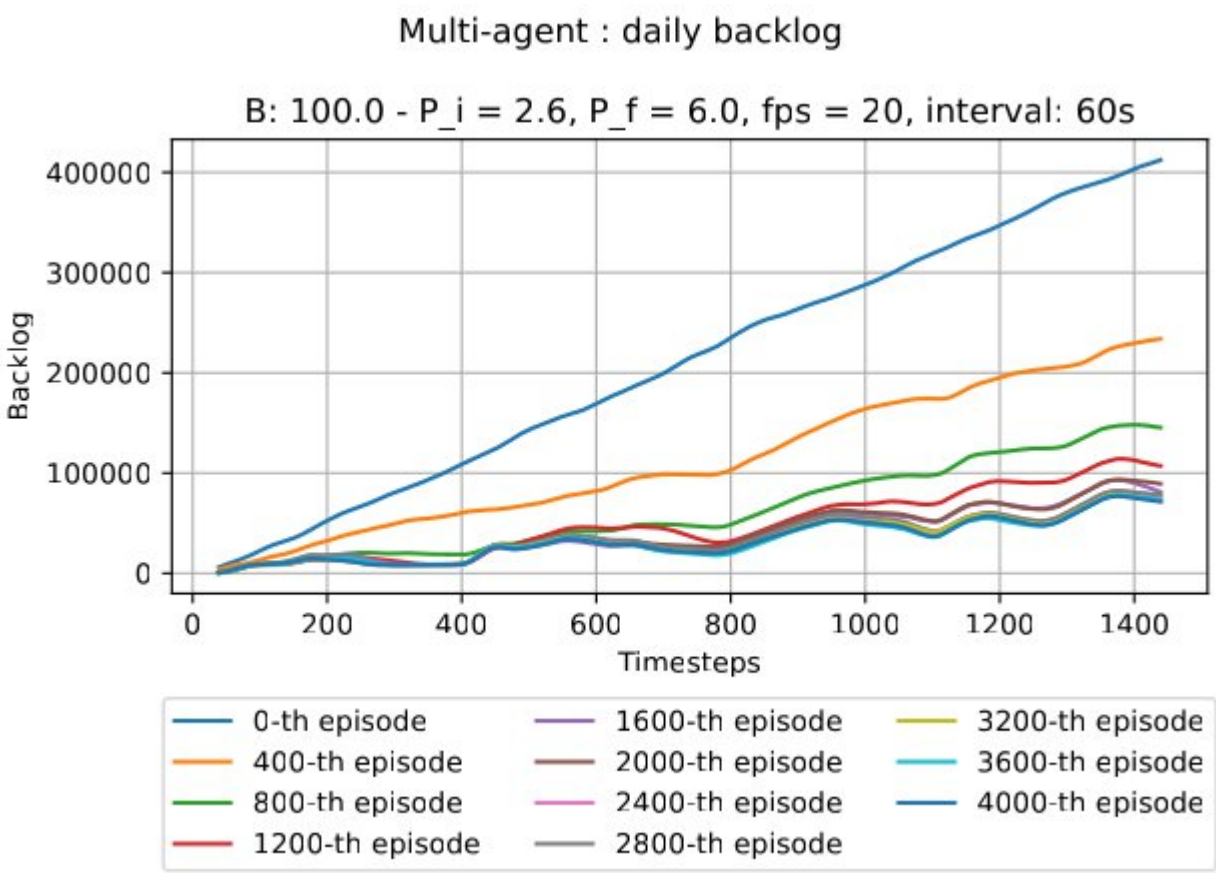


DQN

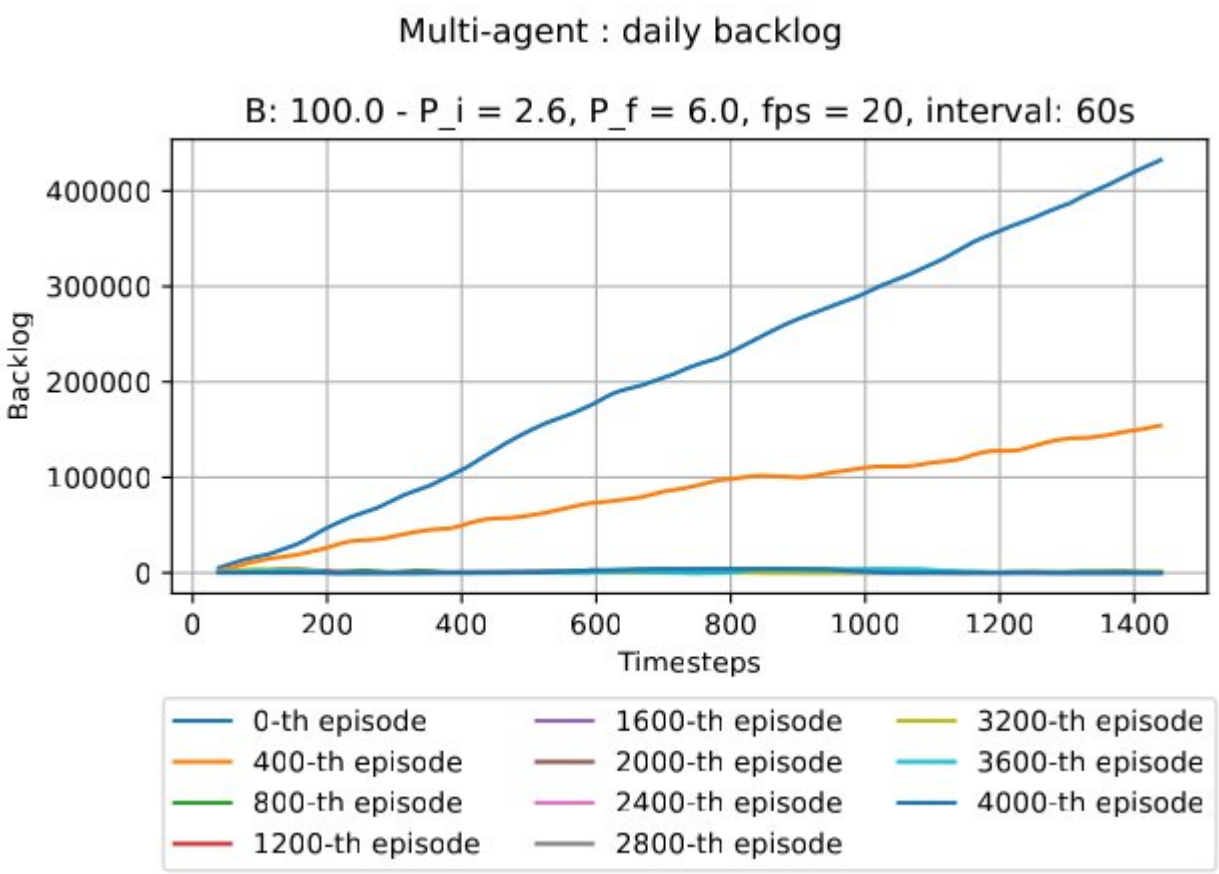


Day 355: backlog 100Wh

TABULAR

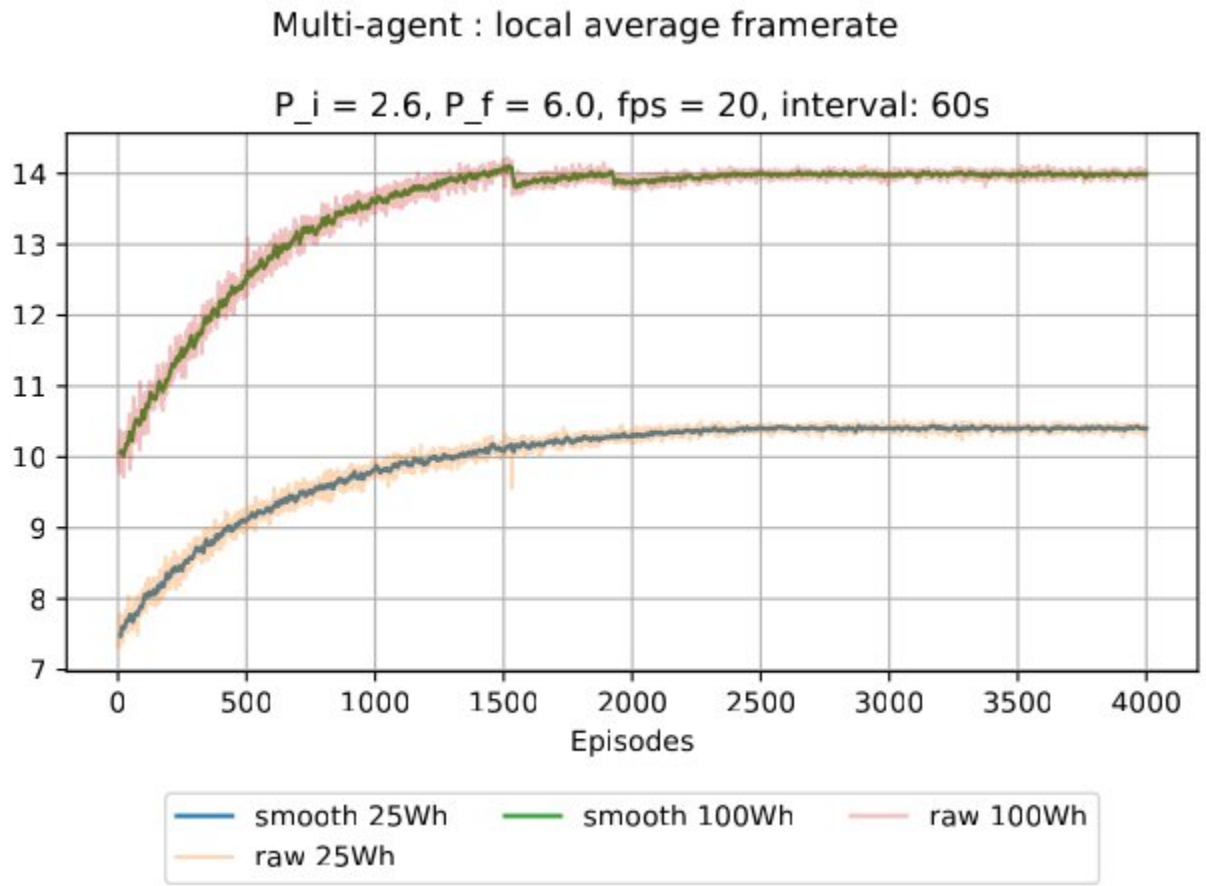


DQN

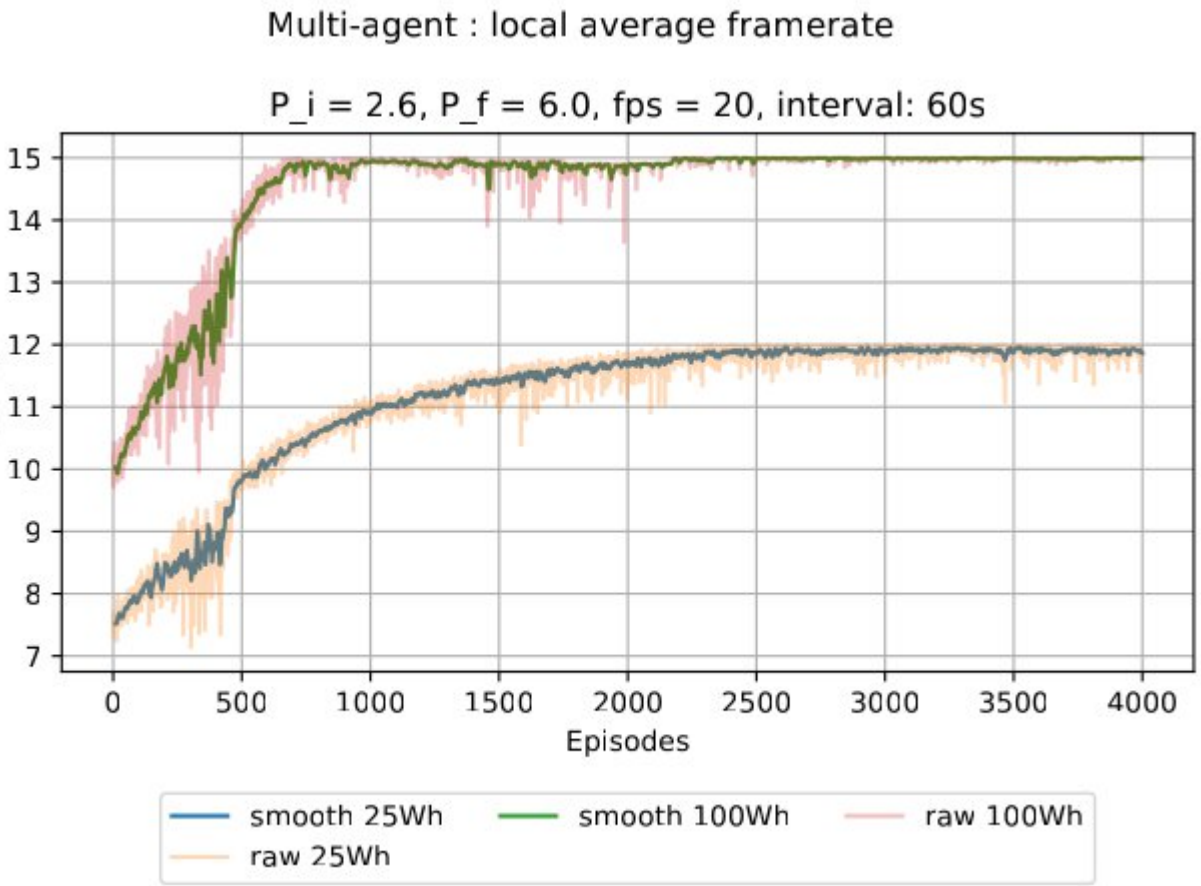


Day 355: local framerates

TABULAR

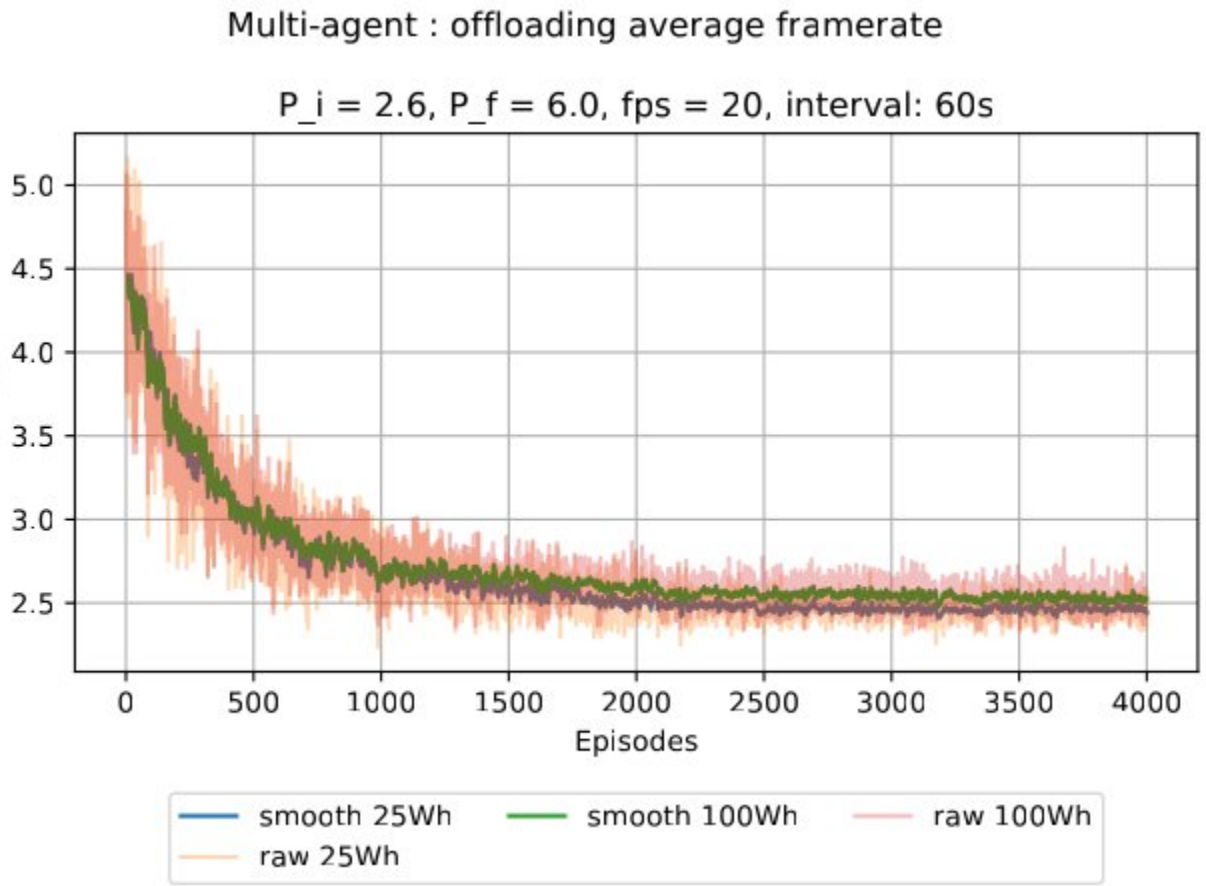


DQN

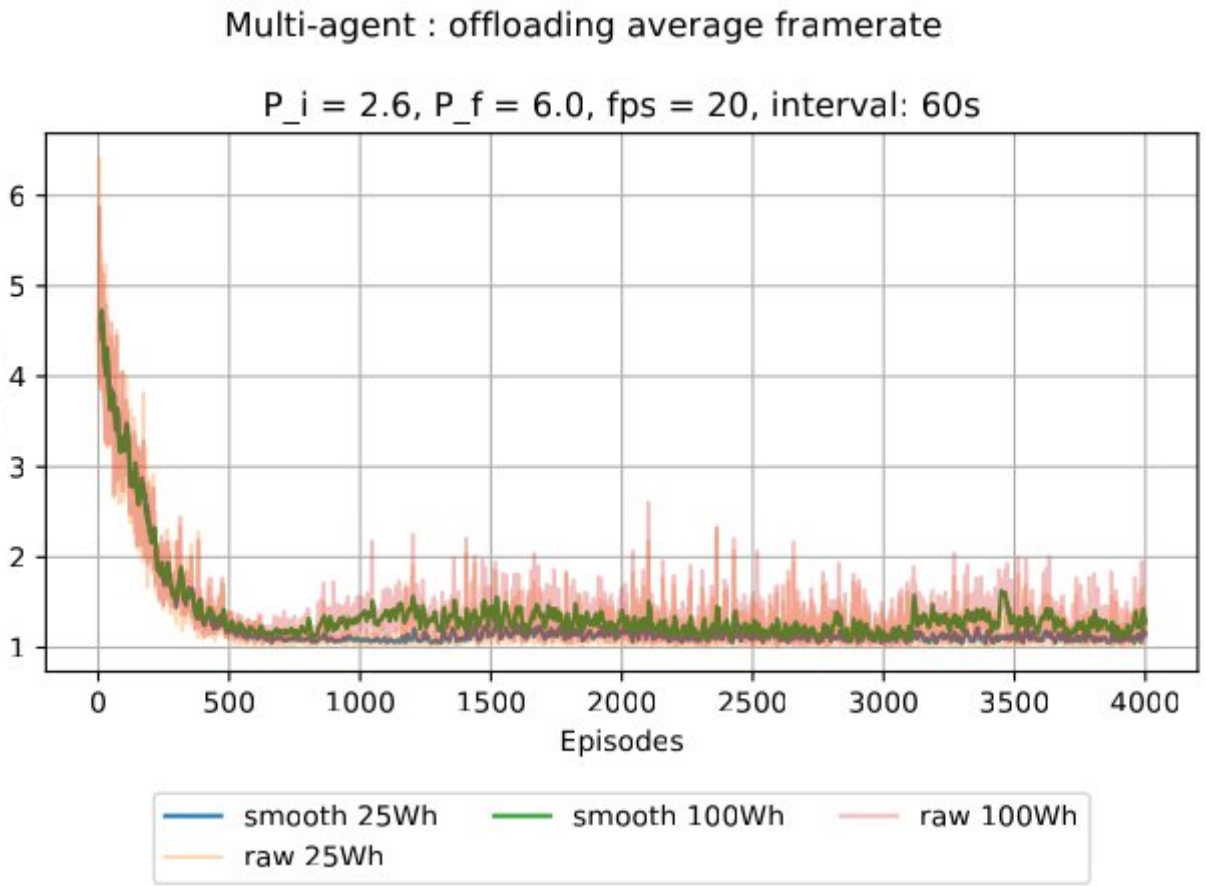


Day 355: offloading framerates

TABULAR

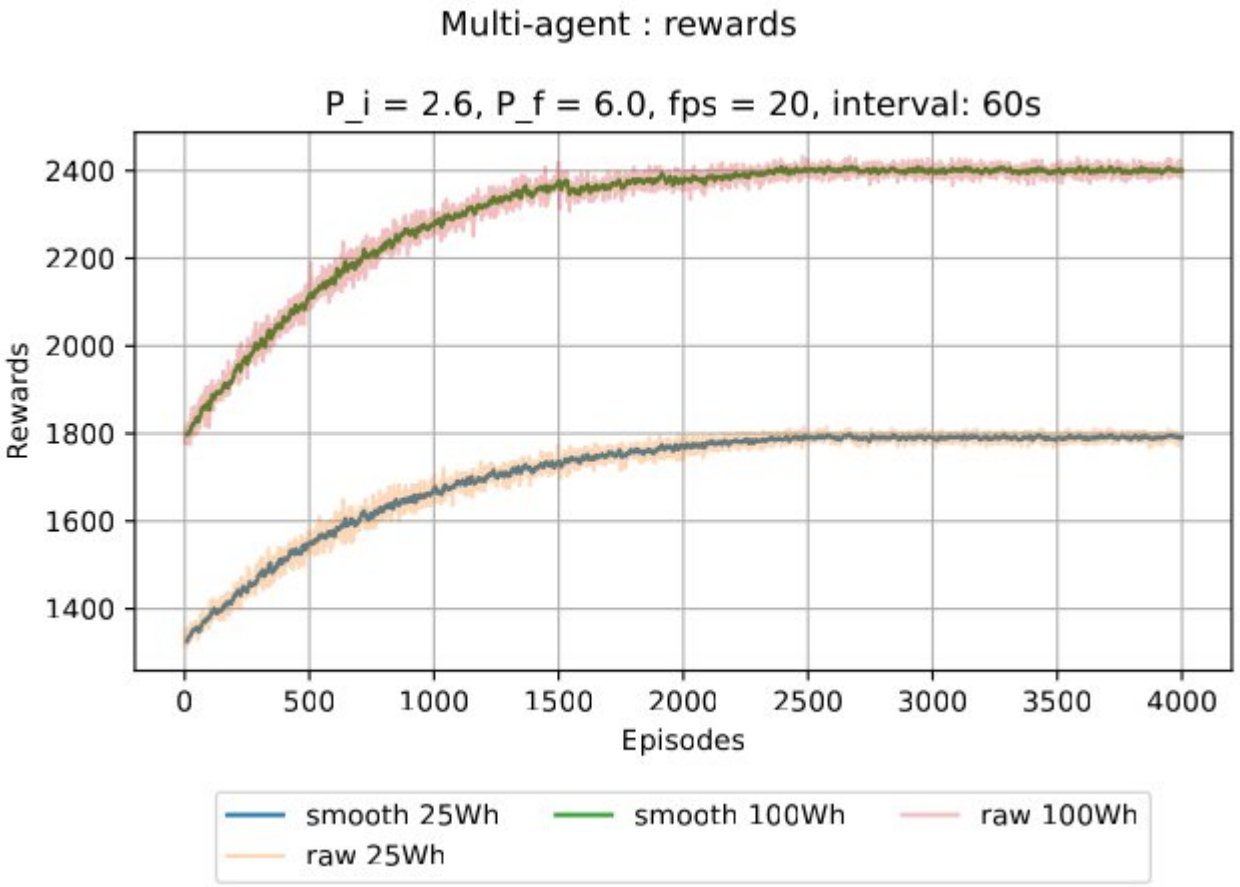


DQN



Day 355: rewards

TABULAR



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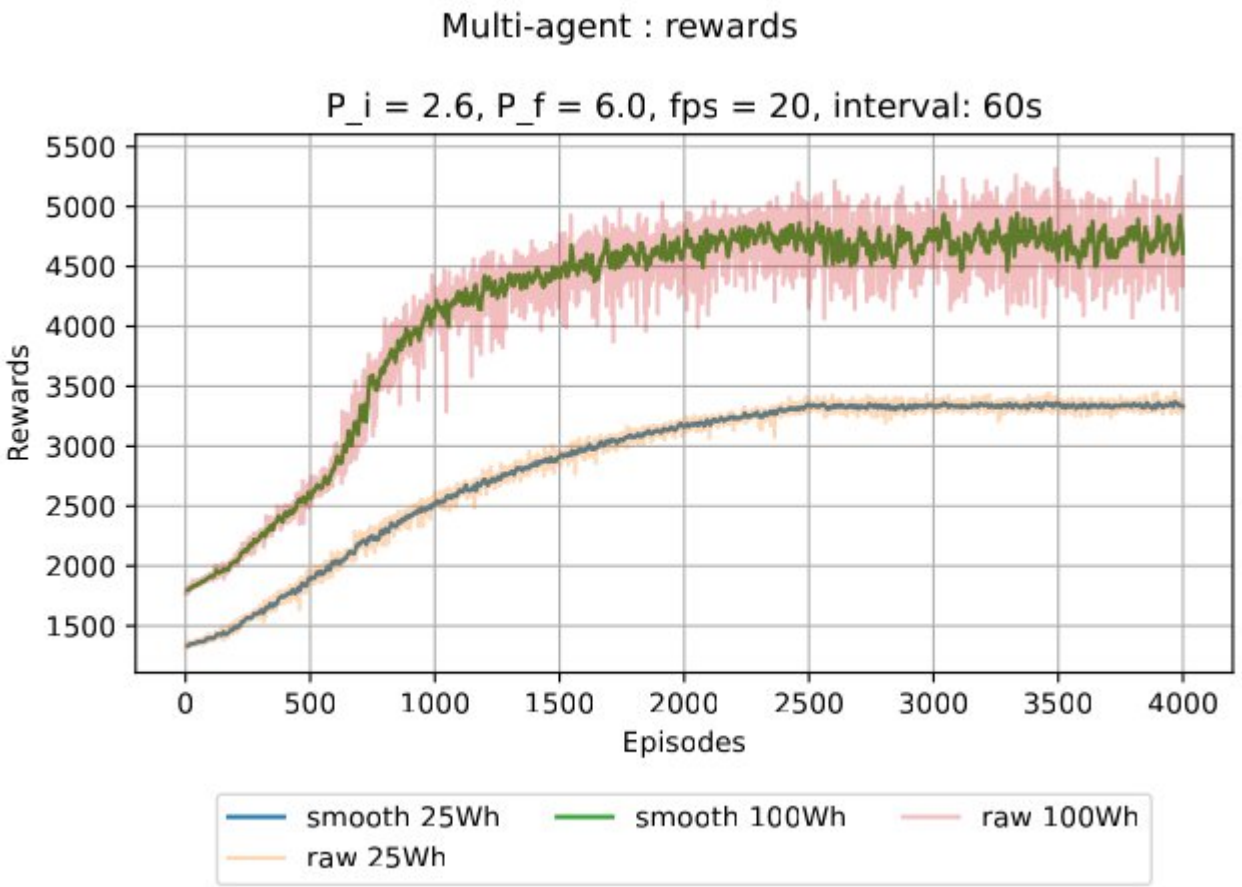
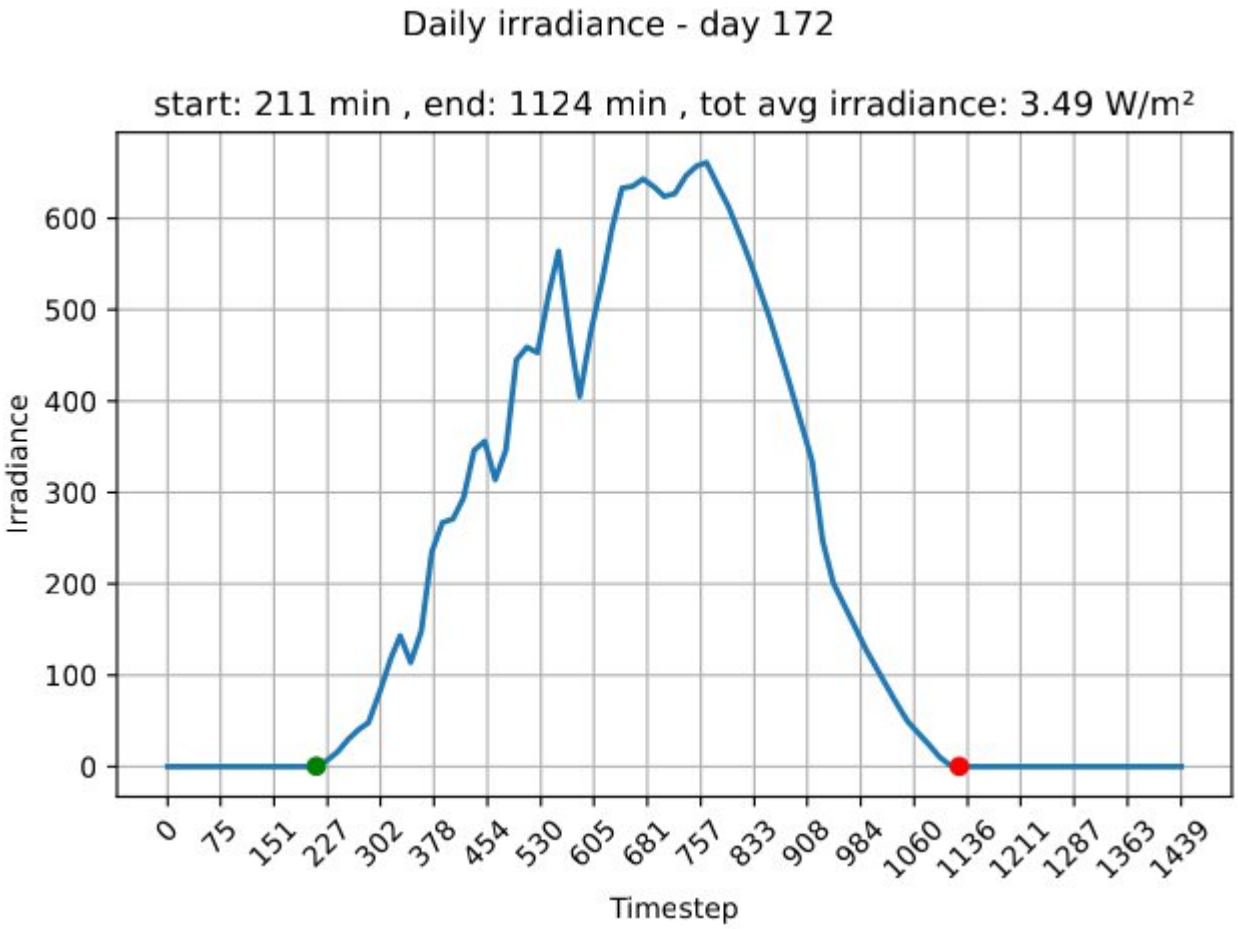
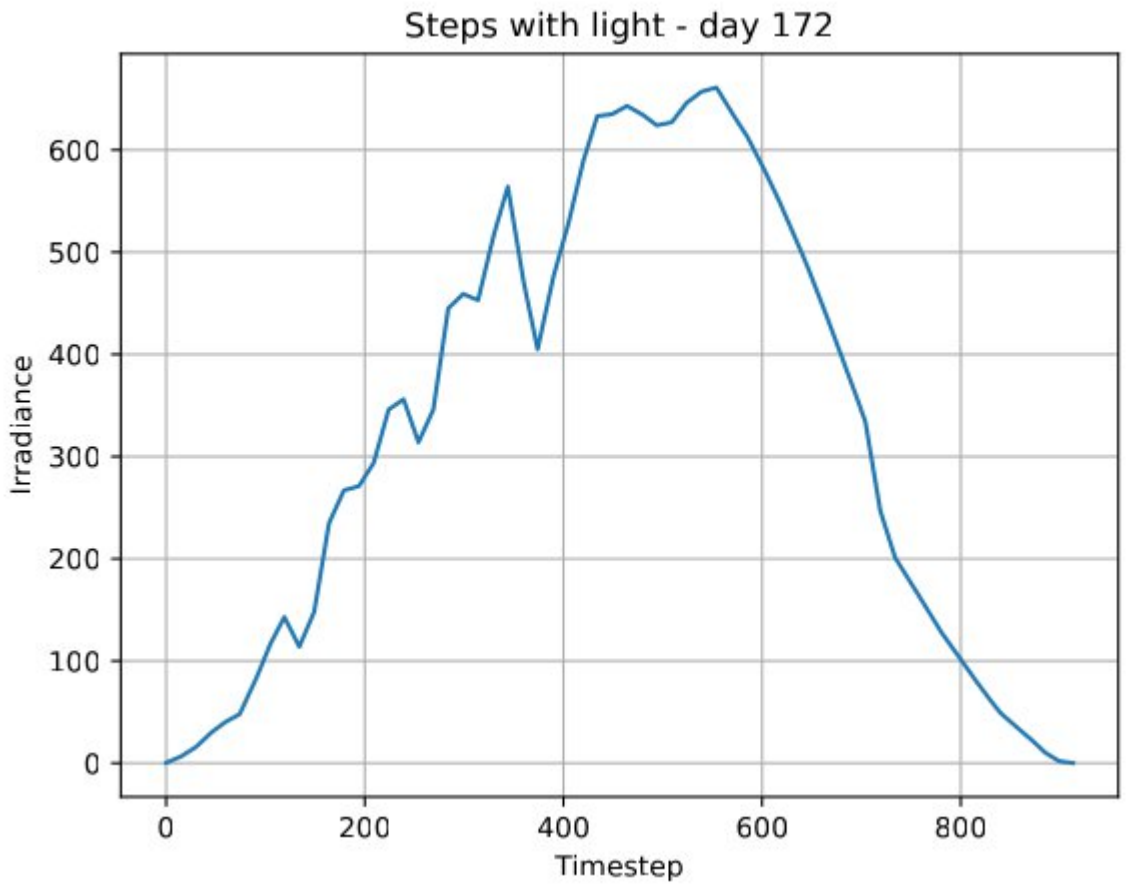


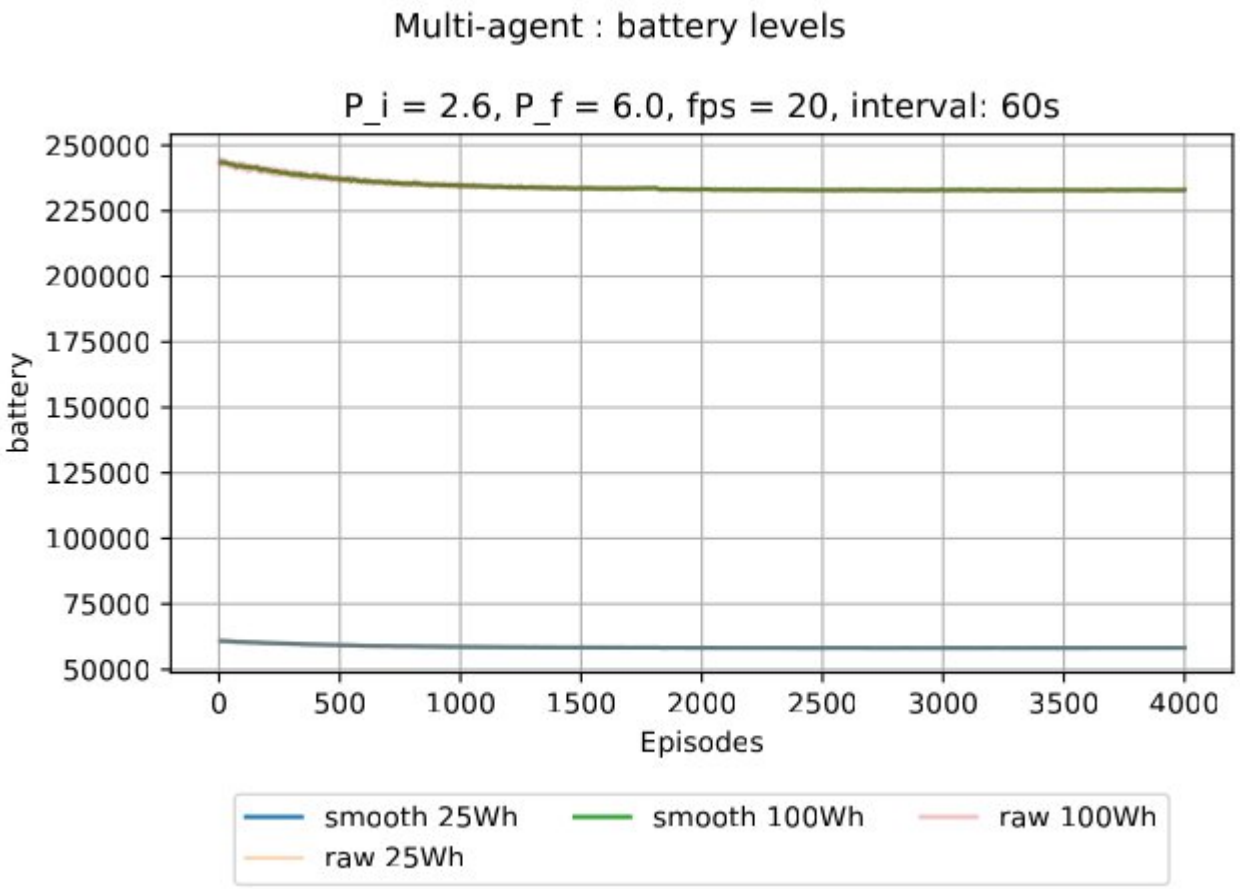
Day 172


Daily irradiance - day 172
start: 211 min , end: 1124 min , tot avg irradiance: 3.49 W/m²
Irradiance
0
100
200
300
400
500
600
0
75
151
227
302
378
454
530
605
681
757
833
908
984
1060
1136
1211
1287
1363
1439
Timestep

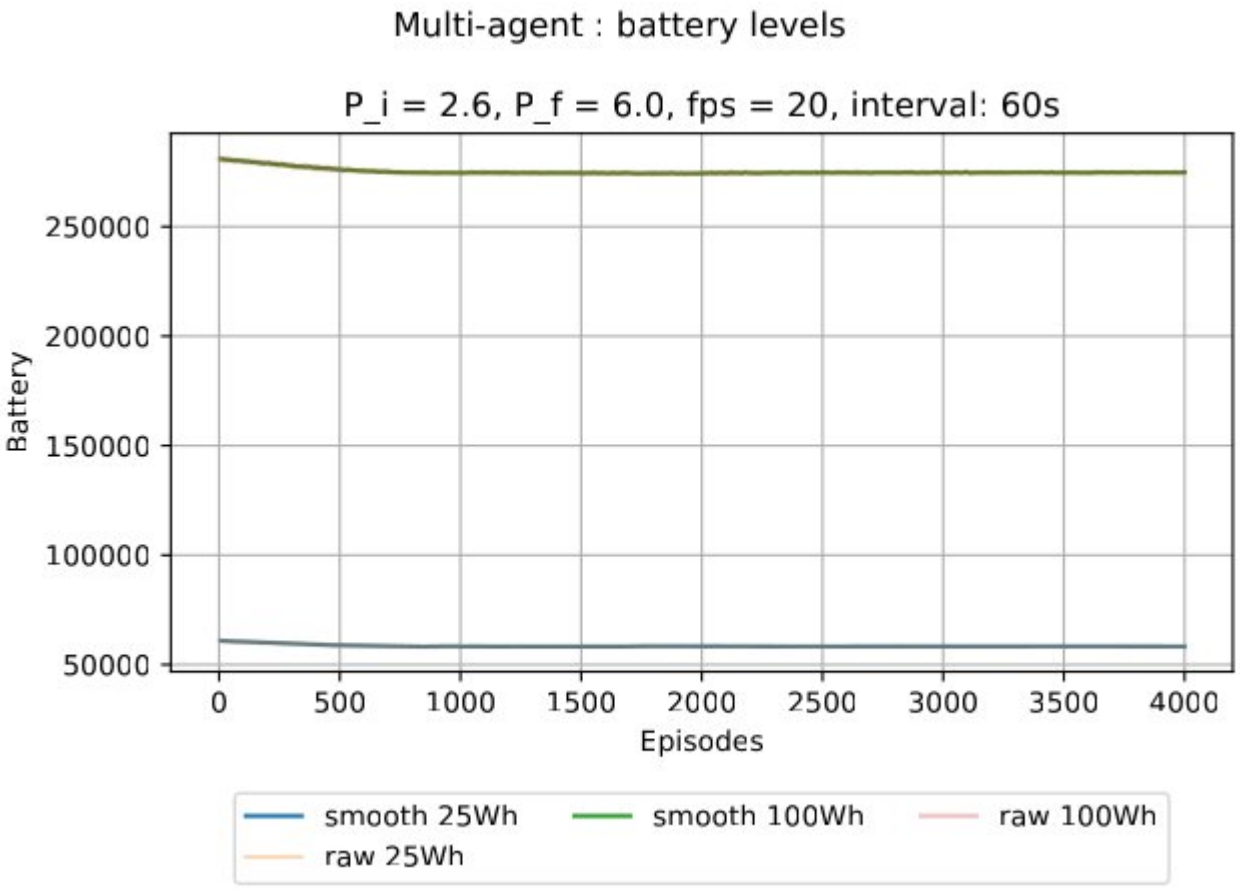

Steps with light - day 172
Irradiance
0
100
200
300
400
500
600
0
200
400
600
800
Timestep

Day 172: batteries

TABULAR

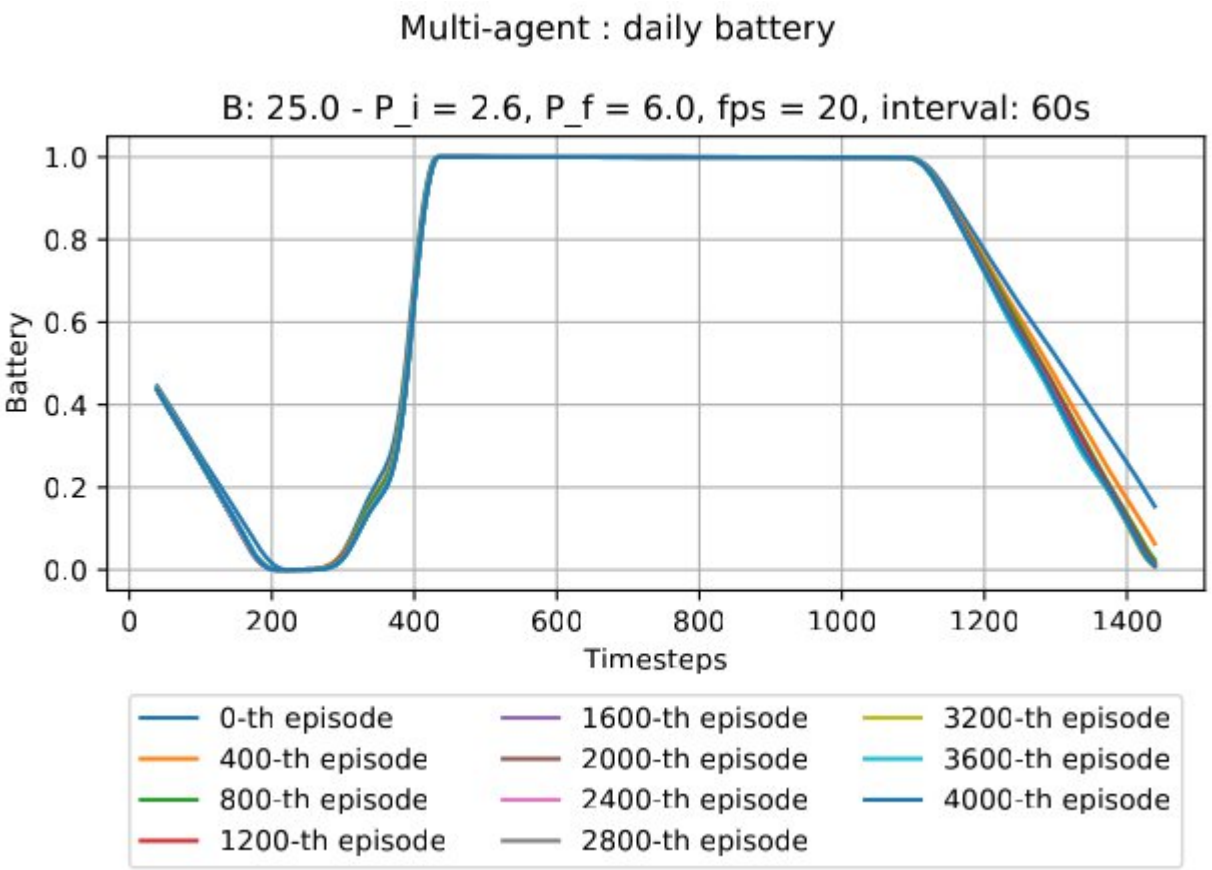


DQN

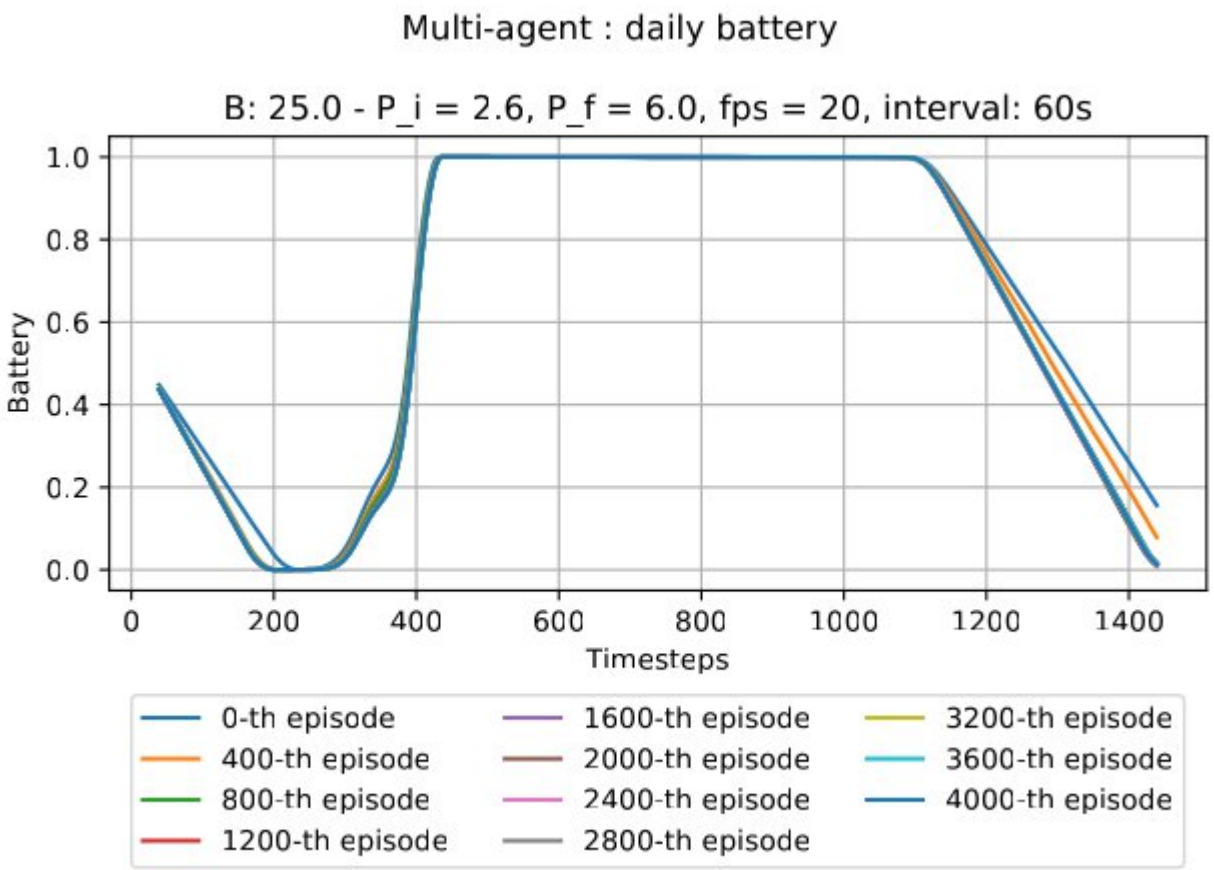


Day 172: battery 25Wh

TABULAR

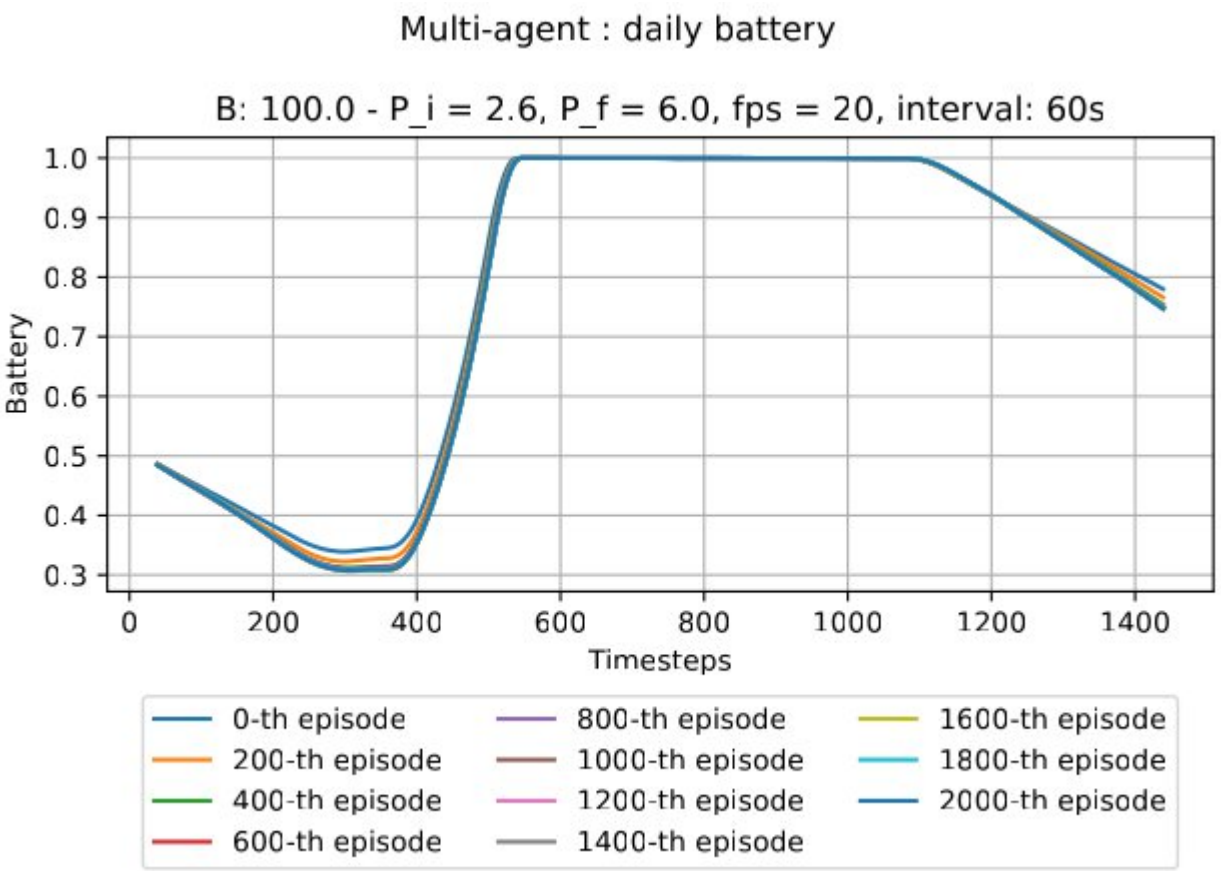


DQN



Day 172: battery 100Wh

TABULAR



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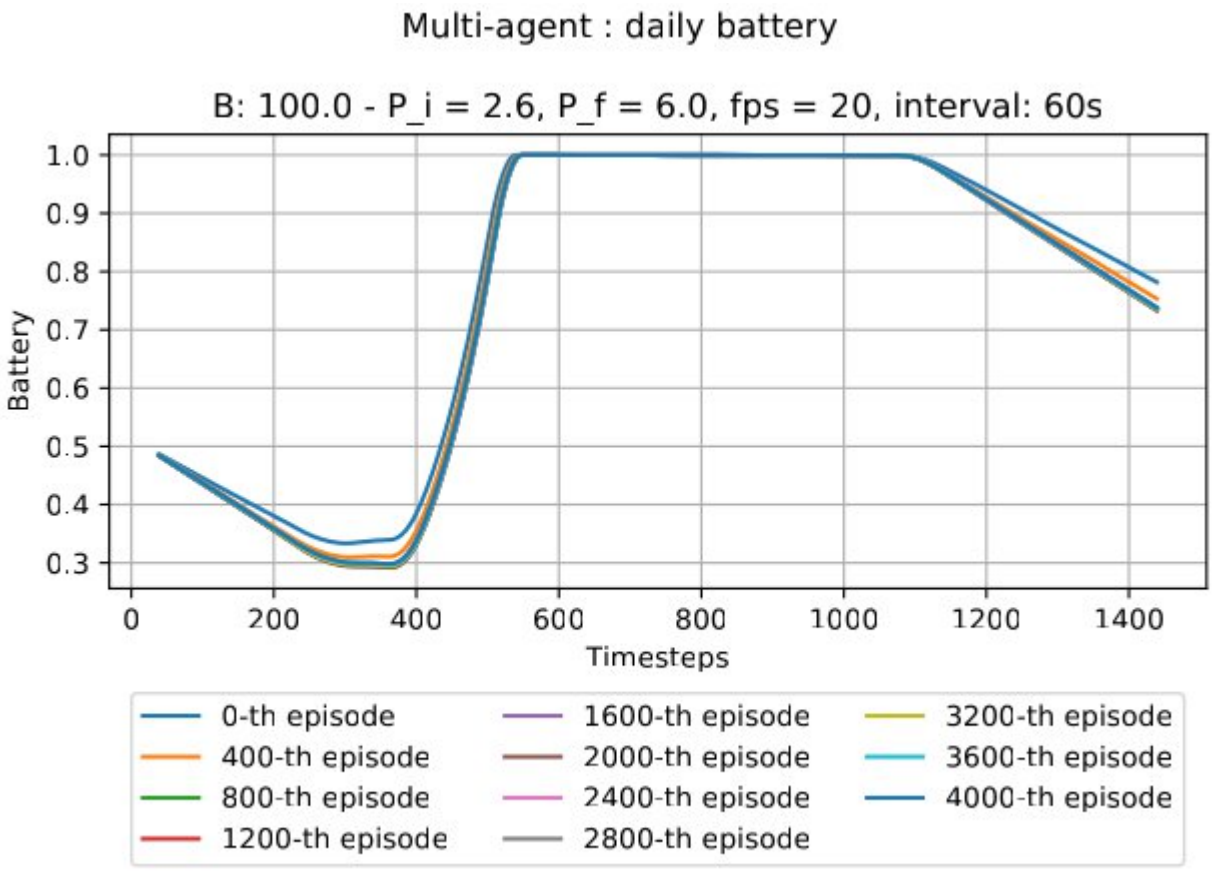
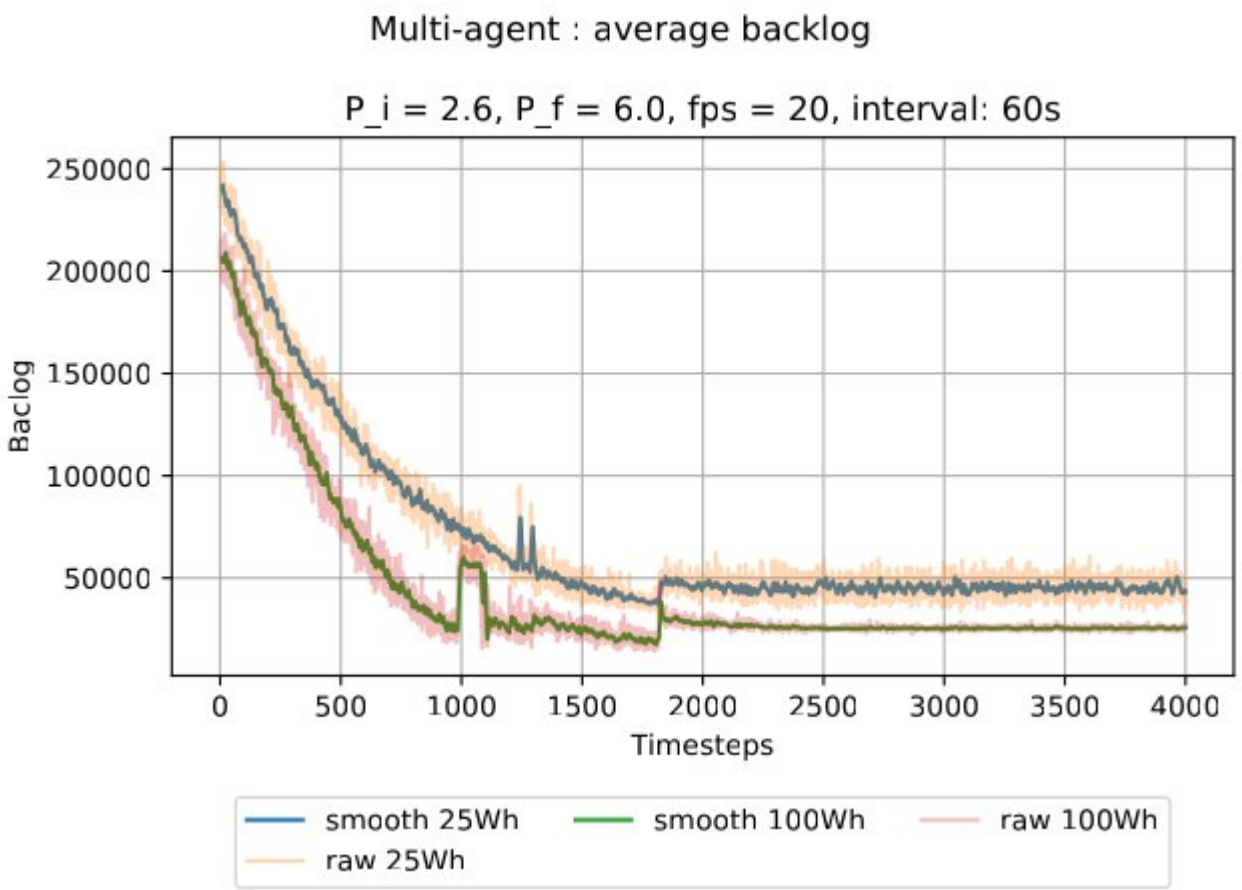
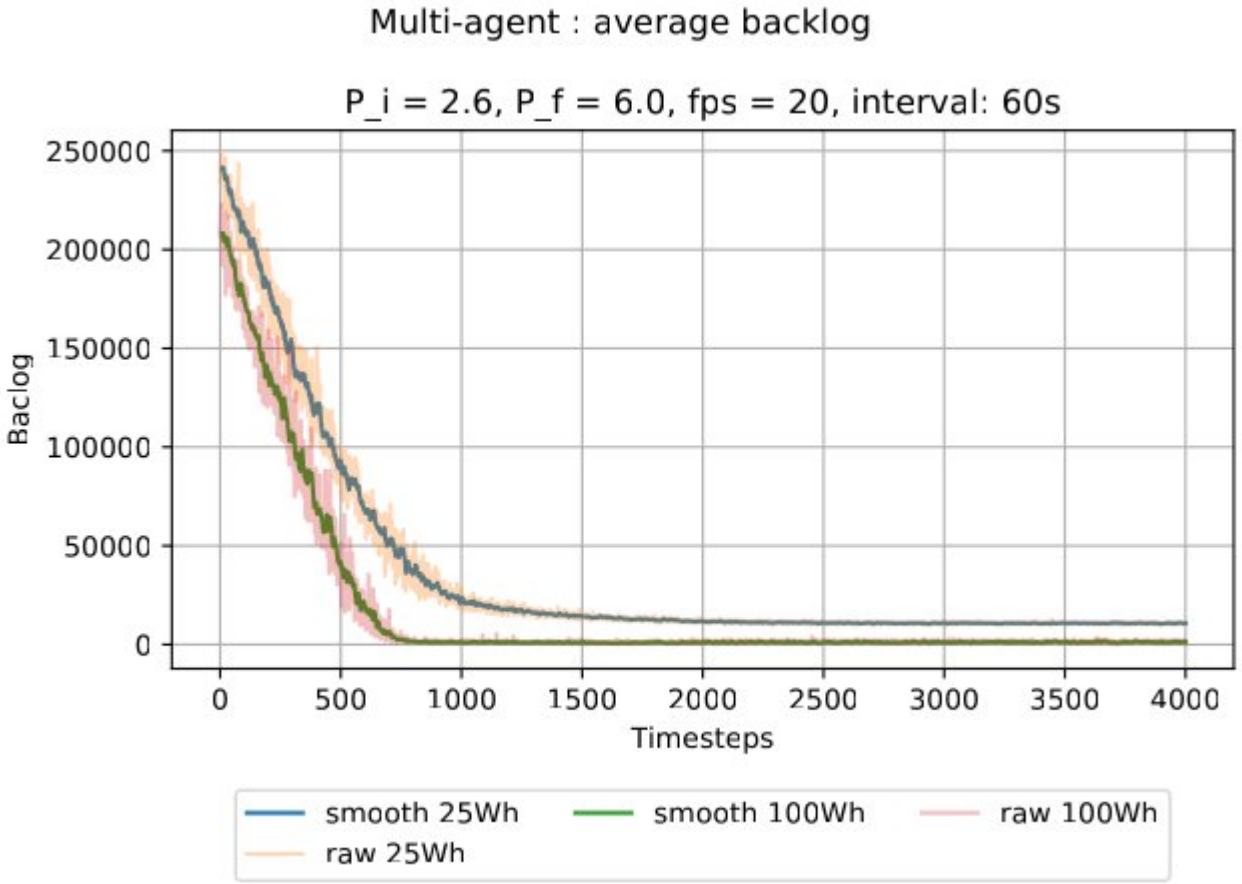


Day 172: backlogs

TABULAR

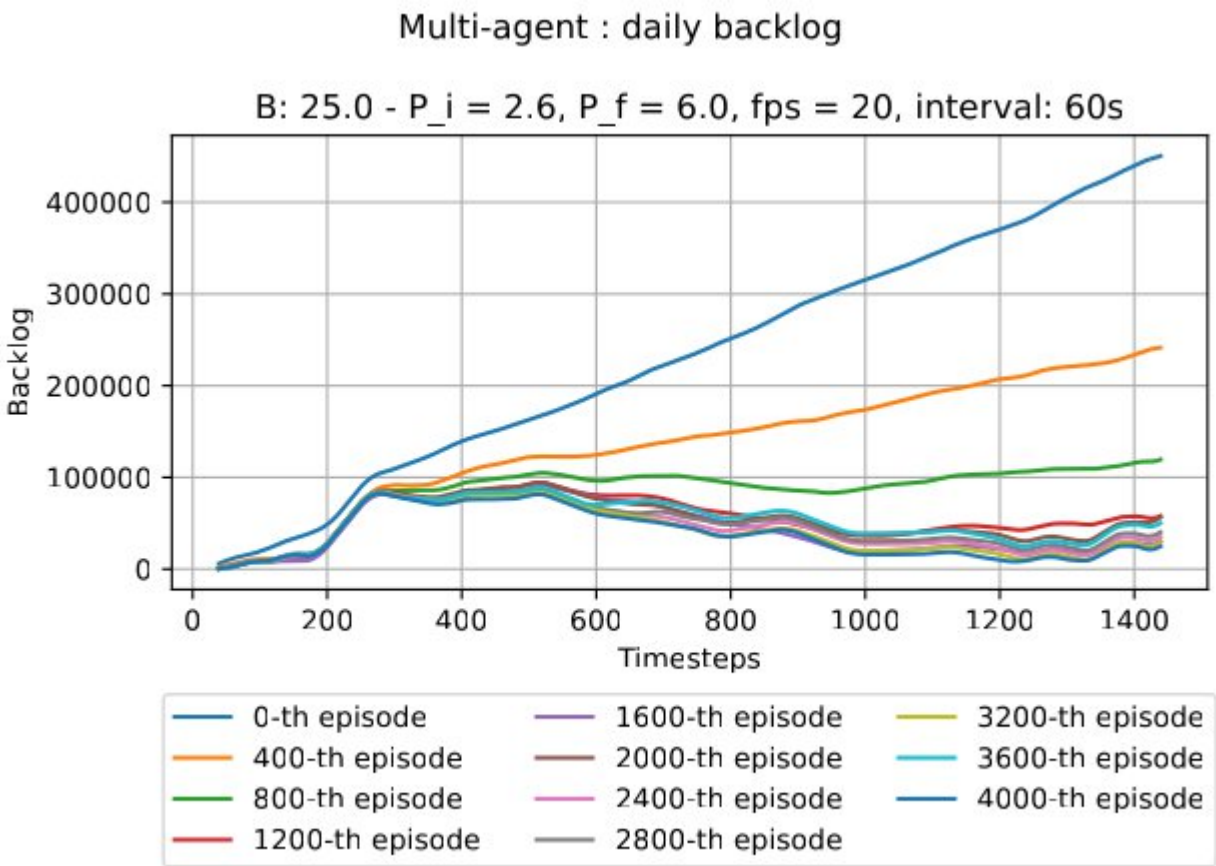


DQN

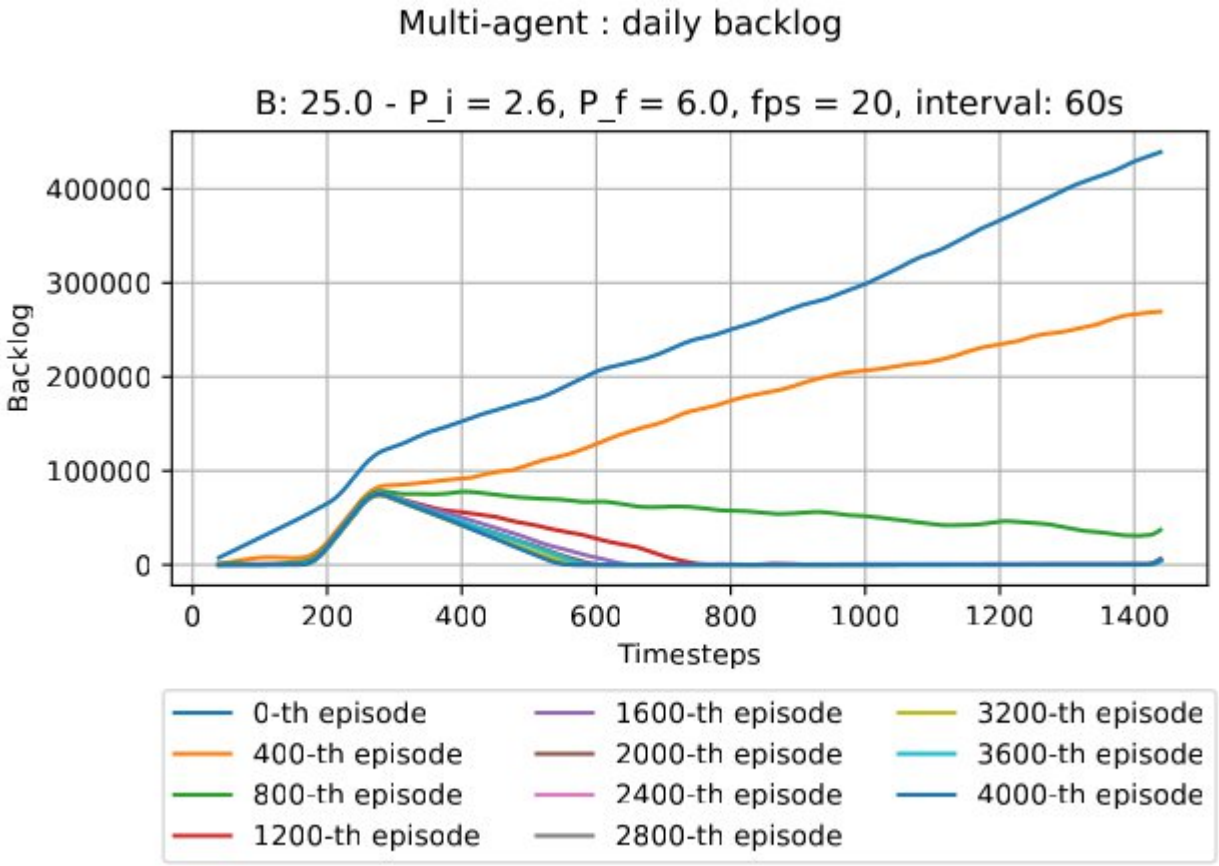


Day 172: backlog 25Wh

TABULAR

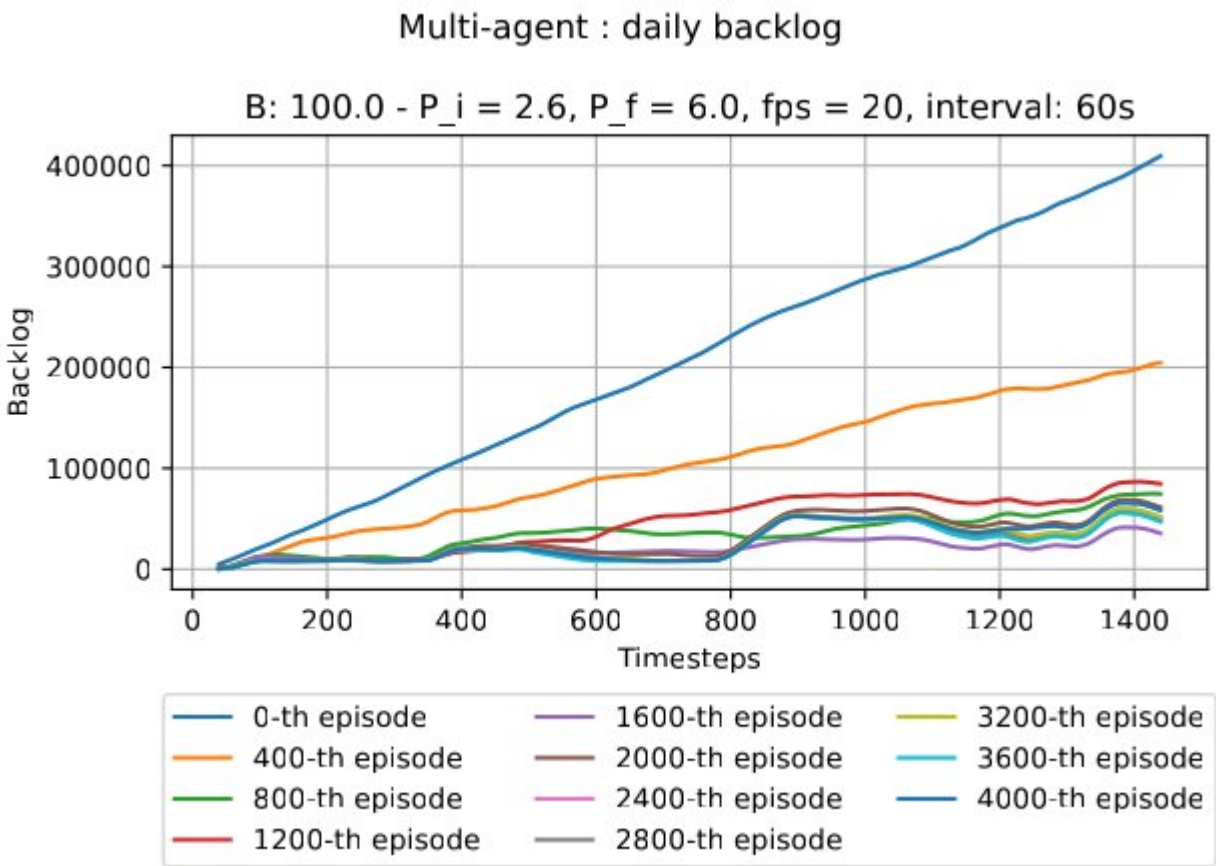


DQN



Day 172: backlog 100Wh

TABULAR



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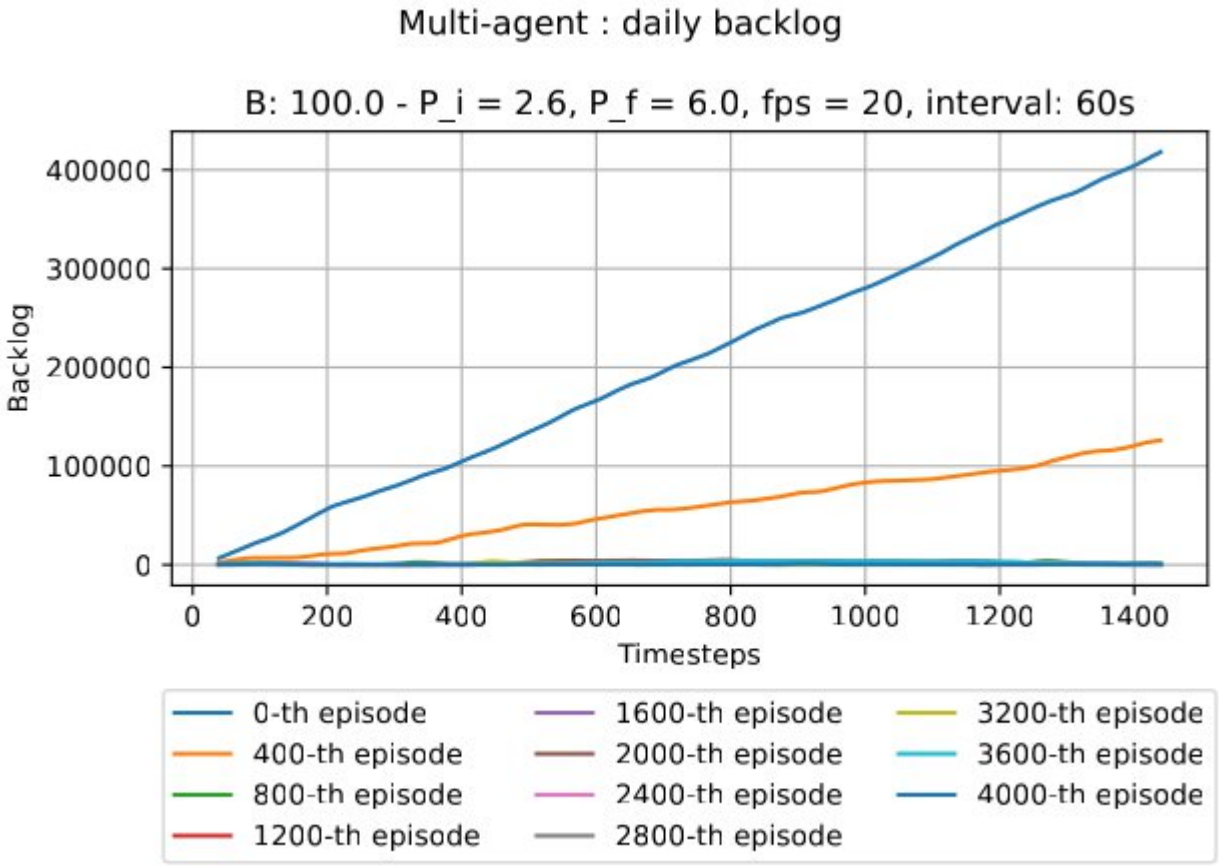
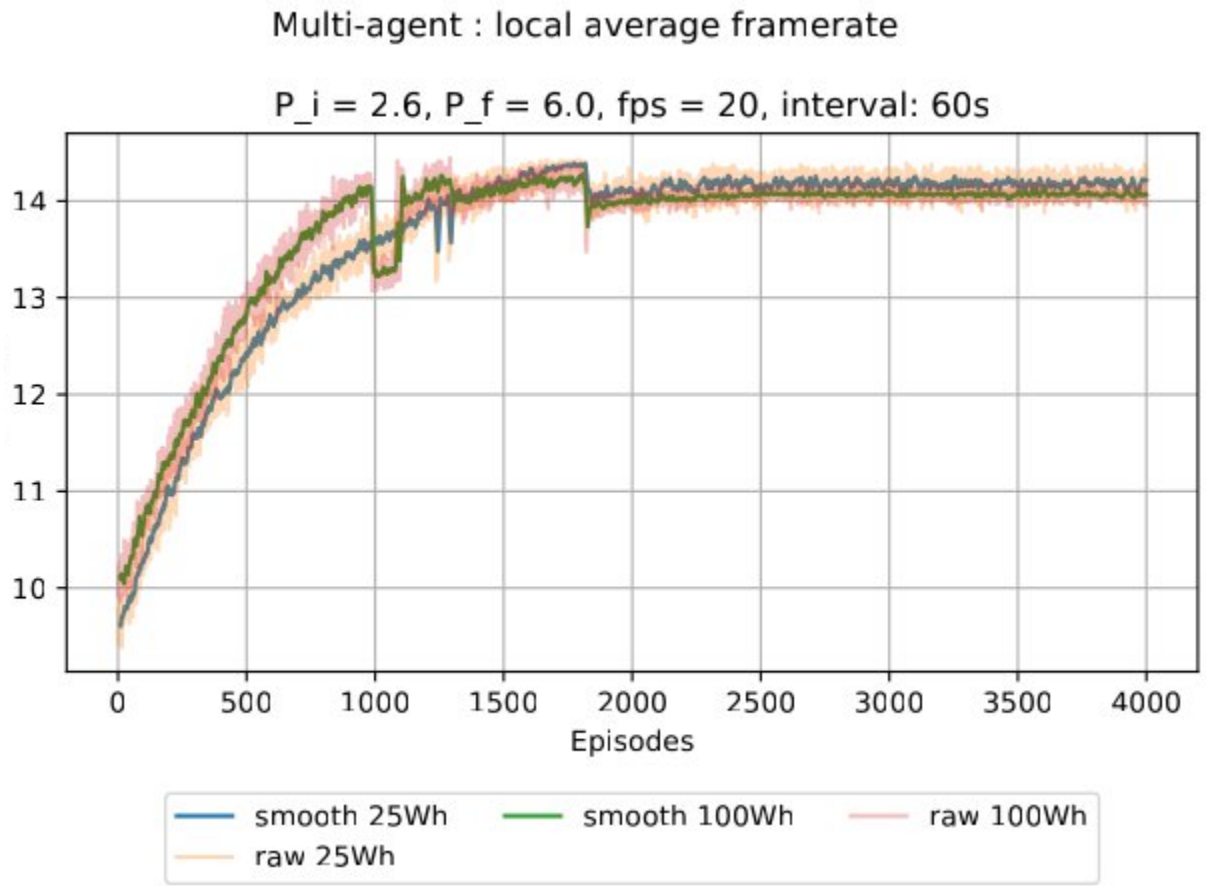


Day 172: local framerates

TABULAR



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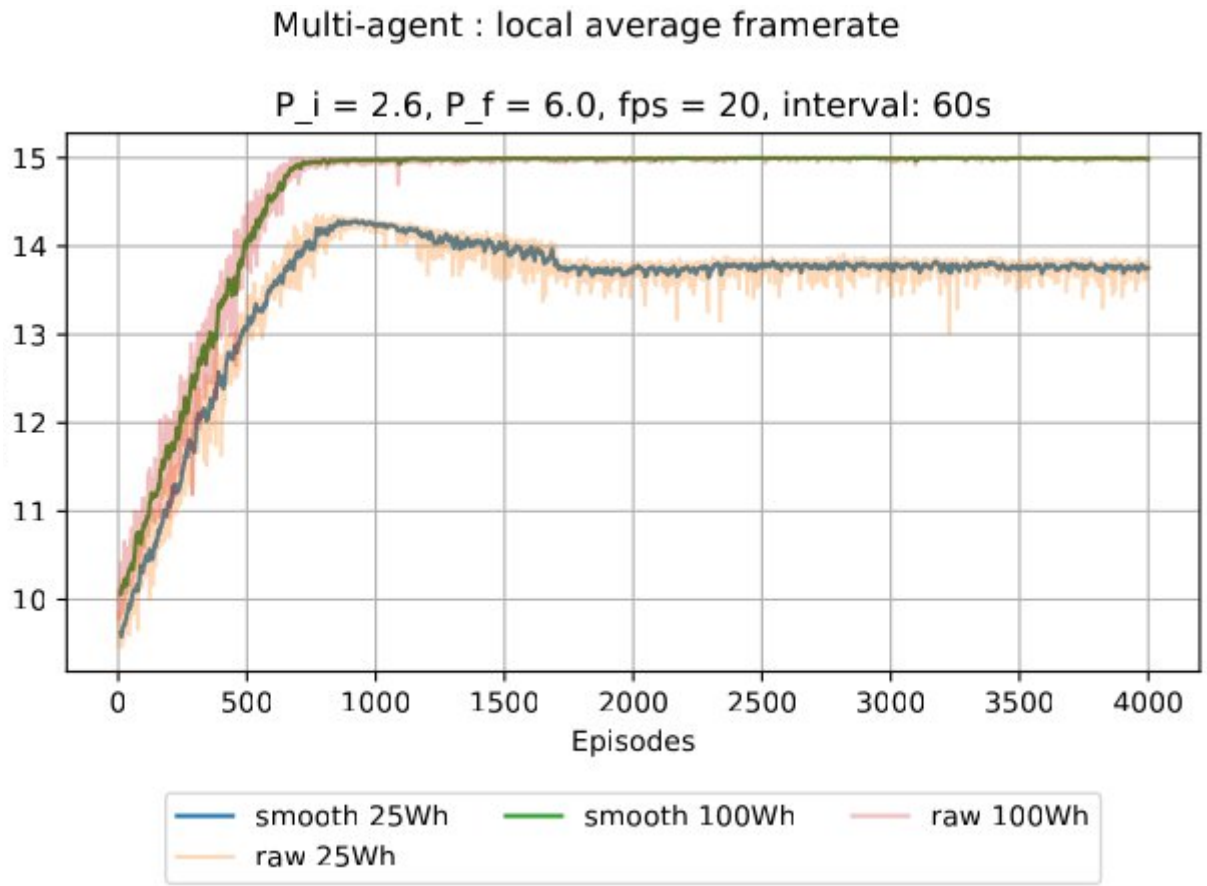
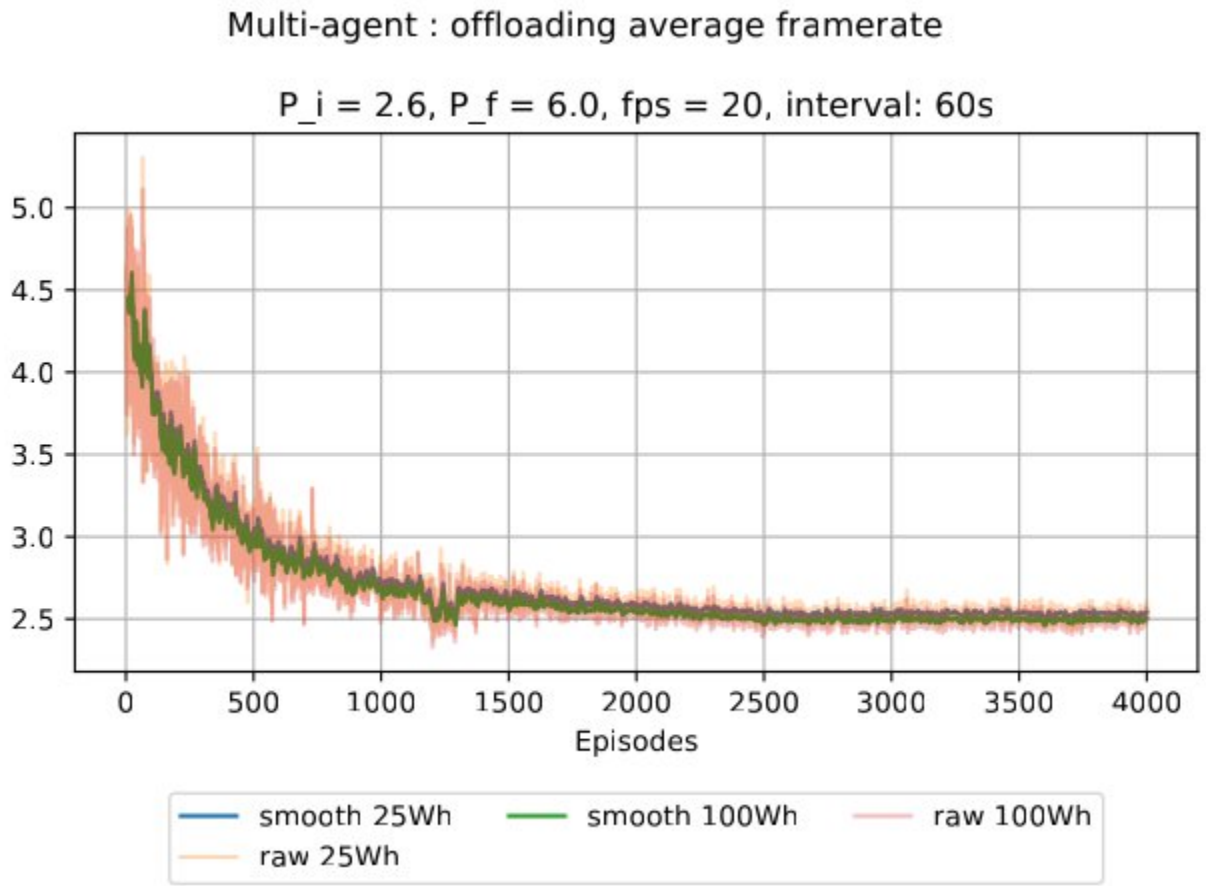


Day 172: offloading framerates

TABULAR



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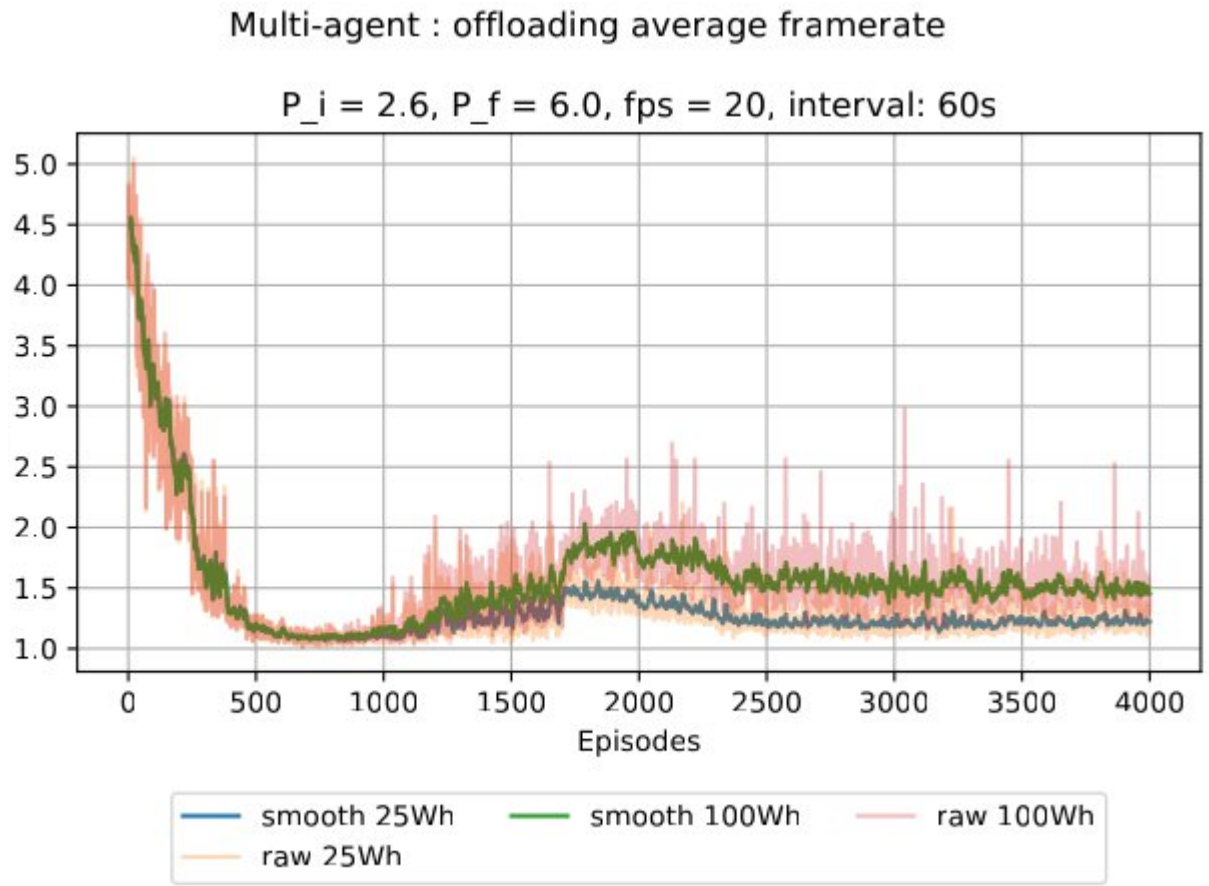
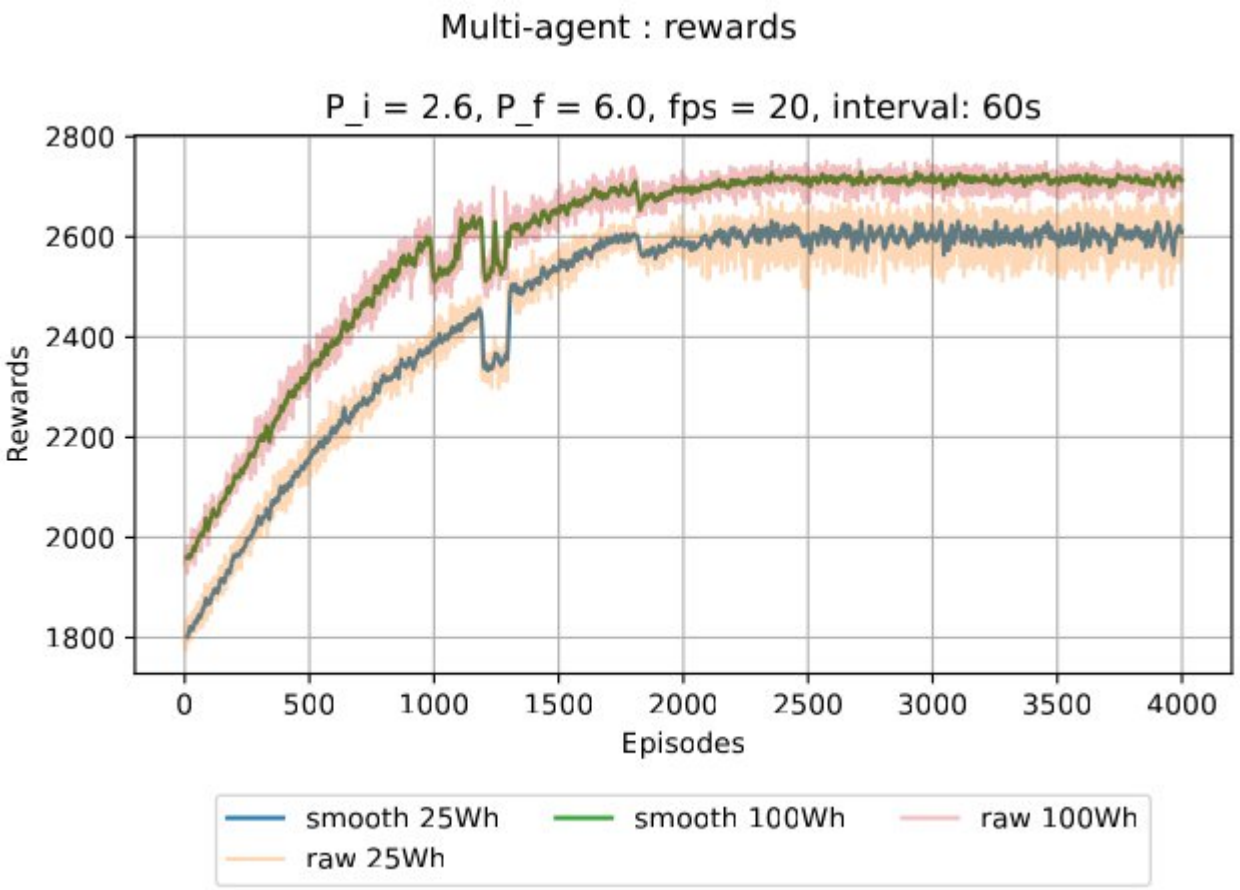
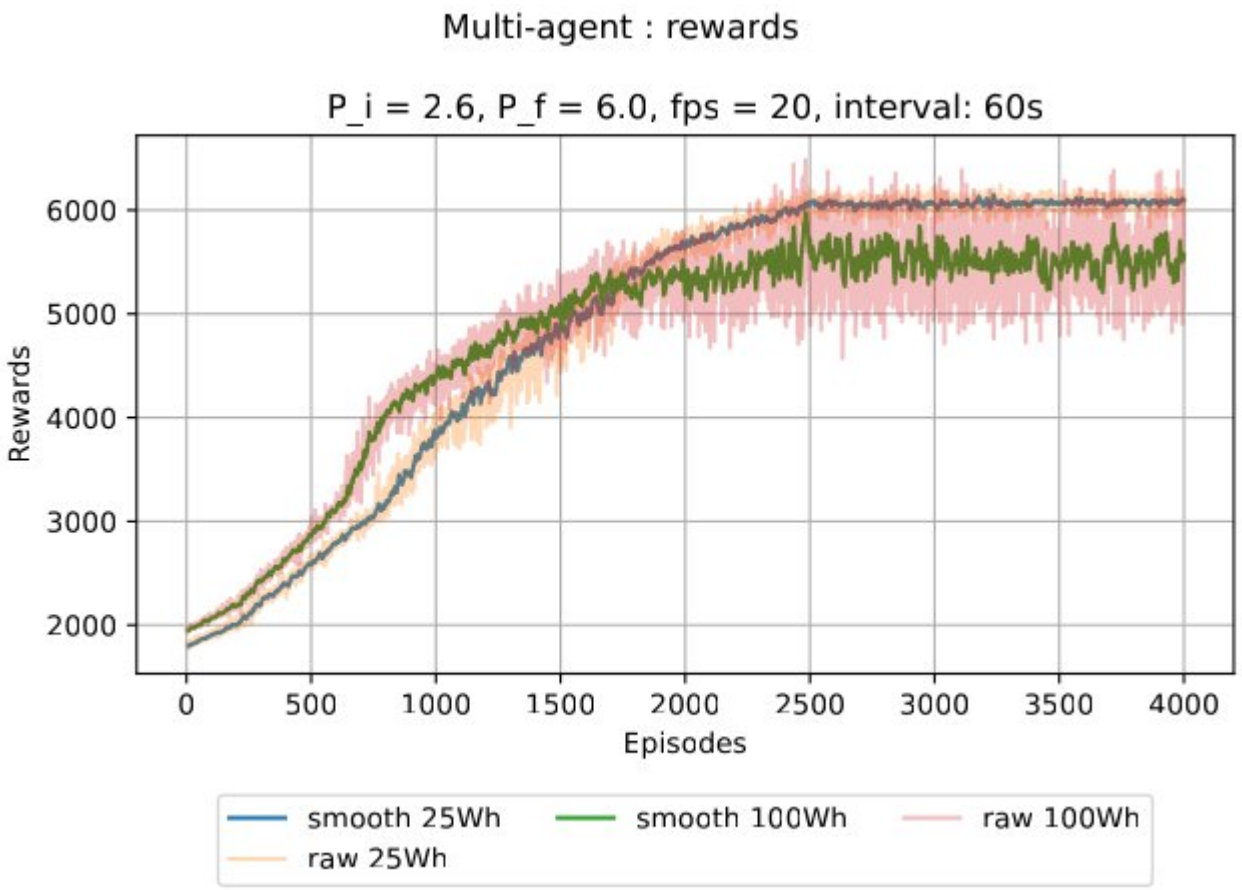


Day 172: rewards

TABULAR



DQN